Table 14-1 DUCT MATERIAL ROUGHNESS FACTORS

Duct Material	Roughness Category	Absolute Roughness ϵ, ft	Absolute Roughness ϵ, mm
Uncoated carbon steel, clean (Moody 1944) (0.00015 ft) (0.05 mm) PVC plastic pipe (Swim 1982) (0.0003 to 0.00015 ft) (0.01 to 0.05 mm) Aluminum (Hutchinson 1953) (0.00015 to 0.0002 ft) (0.04 to 0.06 mm)	Smooth	0.0001	0.03
Galvanized steel, longitudinal seams, 4 ft (1200 mm) joints (Griggs 1987) (0.00016 to 0.00032 ft) (0.05 to 0.1 mm)	Medium Smooth	0.0003	0.09
Galvanized steel, spiral seam with 1, 2, and 3 ribs, 12 ft (3600 mm) joints (Jones 1979, Griggs 1987) (0.00018 to 0.00038 ft) (0.05 to 0.12 mm)	(New Duct Friction Loss Chart)		
Hot-dipped galvanized steel, longitudinal seams, 2.5 ft (760 mm) joints (Wright 1945) (0.0005 ft) (0.15 mm)	Old Average	0.0005	0.15
Fibrous glass duct, rigid Fibrous glass duct liner, air side with facing material (Swim 1978) (0.005 ft) (1.5 mm)	Medium rough	0.003	0.9
Fibrous glass duct liner, air side spray coated (Swim 1978) (0.015 ft) (4.5 mm) Flexible duct, metallic, (0.004 to 0.007 ft (1.2 to 2.1 mm) when fully extended) Flexible duct, all types of fabric and wire (0.0035 to 0.015 ft (1.0 to 4.6 mm) when fully extended) Concrete (Moody 1944) (0.001 to 0.01 ft) (0.3 to 3.0 mm)	Rough	0.01	3.0

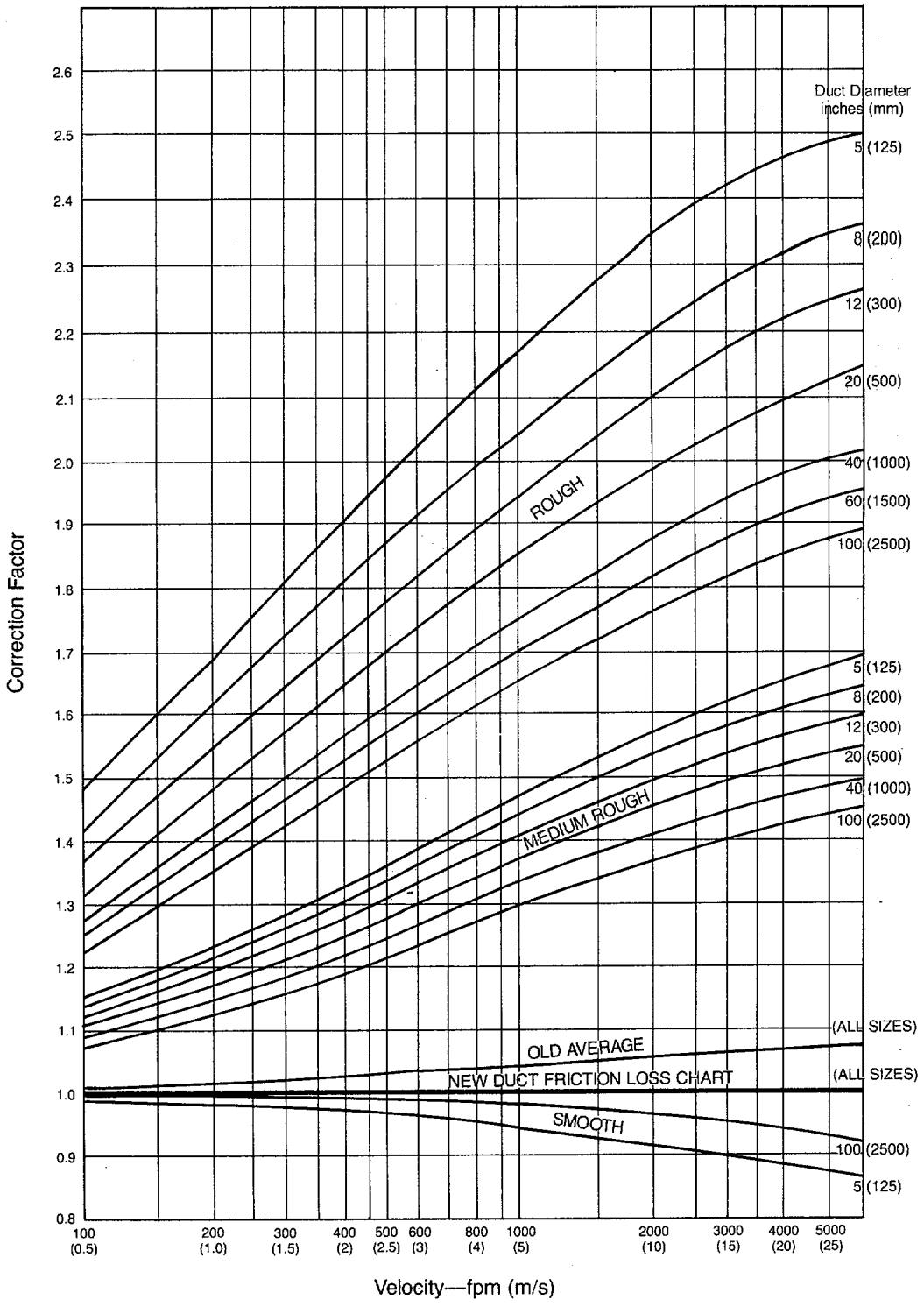


Figure 14-3 DUCT FRICTION LOSS CORRECTION FACTORS

Table 14-2 CIRCULAR EQUIVALENTS OF RECTANGULAR DUCTS FOR EQUAL FRICTION AND CAPACITY (U.S. Units) (2)

Dimensions in Inches

Side Rectangular Duct	4.0	4.5	5.0	5.5	6.0	6.5	7.0	7.5	8.0	9.0	10.0	11.0	12.0	13.0	14.0	15.0	16.0
3.0	3.8	4.0	4.2	4.4	4.6	4.7	4.9	5.1	5.2	5.5	5.7	6.0	6.2	6.4	6.6	6.8	7.0
3.5	4.1	4.3	4.6	4.8	5.0	5.2	5.3	5.5	5.7	6.0	6.3	6.5	6.8	7.0	7.2	7.5	7.7
4.0	4.4	4.6	4.9	5.1	5.3	5.5	5.7	5.9	6.1	6.4	6.7	7.0	7.3	7.6	7.8	8.1	8.3
4.5	4.6	4.9	5.2	5.4	5.7	5.9	6.1	6.3	6.5	6.9	7.2	7.5	7.8	8.1	8.4	8.6	8.8
5.0	4.9	5.2	5.5	5.7	6.0	6.2	6.4	6.7	6.9	7.3	7.6	8.0	8.3	8.6	8.9	9.1	9.4
5.5	5.1	5.4	5.7	6.0	6.3	6.5	6.8	7.0	7.2	7.6	8.0	8.4	8.7	9.0	9.3	9.6	9.9

Side Rectangular Duct	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	22	24	26	28	30	Side Rectangular Duct
6	6.6																				6
7	7.1	7.7																			7
8	7.6	8.2	8.7																		8
9	8.0	8.7	9.3	9.8																	9
10	8.4	9.1	9.8	10.4	10.9																10
11	8.8	9.5	10.2	10.9	11.5	12.0															11
12	9.1	9.9	10.7	11.3	12.0	12.6	13.1														12
13	9.5	10.3	11.1	11.8	12.4	13.1	13.7	14.2													13
14	9.8	10.7	11.5	12.2	12.9	13.5	14.2	14.7	15.3												14
15	10.1	11.0	11.8	12.6	13.3	14.0	14.6	15.3	15.8	16.4											15
16	10.4	11.3	12.2	13.0	13.7	14.4	15.1	15.7	16.4	16.9	17.5										16
17	10.7	11.6	12.5	13.4	14.1	14.9	15.6	16.2	16.8	17.4	18.0	18.6									17
18	11.0	11.9	12.9	13.7	14.5	15.3	16.0	16.7	17.3	17.9	18.5	19.1	19.7								18
19	11.2	12.2	13.2	14.1	14.9	15.7	16.4	17.1	17.8	18.4	19.0	19.6	20.2	20.8							19
20	11.5	12.5	13.5	14.4	15.2	16.0	16.8	17.5	18.2	18.9	19.5	20.1	20.7	21.3	21.9						20
22	12.0	13.0	14.1	15.0	15.9	16.8	17.6	18.3	19.1	19.8	20.4	21.1	21.7	22.3	22.9	24.0					22
24	12.4	13.5	14.6	15.6	16.5	17.4	18.3	19.1	19.9	20.6	21.3	22.0	22.7	23.3	23.9	25.1	26.2				24
26	12.8	14.0	15.1	16.2	17.1	18.1	19.0	19.8	20.6	21.4	22.1	22.9	23.5	24.2	24.9	26.1	27.3	28.4			26
28	13.2	14.5	15.6	16.7	17.7	18.7	19.6	20.5	21.3	22.1	22.9	23.7	24.4	25.1	25.8	27.1	28.3	29.5	30.6		28
30	13.6	14.9	16.1	17.2	18.3	19.3	20.2	21.1	22.0	22.9	23.7	24.4	25.2	25.9	26.6	28.0	29.3	30.5	31.7	32.8	30
32	14.0	15.3	16.5	17.7	18.8	19.8	20.8	21.8	22.7	23.5	24.4	25.2	26.0	26.7	27.5	28.9	30.2	31.5	32.7	33.9	32
34	14.4	15.7	17.0	18.2	19.3	20.4	21.4	22.4	23.3	24.2	25.1	25.9	26.7	27.5	28.3	29.7	31.0	32.4	33.7	34.9	34
36	14.7	16.1	17.4	18.6	19.8	20.9	21.9	22.9	23.9	24.8	25.7	26.6	27.4	28.2	29.0	30.5	32.0	33.3	34.6	35.9	36
38	15.0	16.5	17.8	19.0	20.2	21.4	22.4	23.5	24.5	25.4	26.4	27.2	28.1	28.9	29.8	31.3	32.8	34.2	35.6	36.8	38
40	15.3	16.8	18.2	19.5	20.7	21.8	22.9	24.0	25.0	26.0	27.0	27.9	28.8	29.6	30.5	32.1	33.6	35.1	36.4	37.8	40
42	15.6	17.1	18.5	19.9	21.1	22.3	23.4	24.5	25.6	26.6	27.6	28.5	29.4	30.3	31.2	32.8	34.4	35.9	37.3	38.7	42
44	15.9	17.5	18.9	20.3	21.5	22.7	23.9	25.0	26.1	27.1	28.1	29.1	30.0	30.9	31.8	33.5	35.1	36.7	38.1	39.5	44
46	16.2	17.8	19.3	20.6	21.9	23.2	24.4	25.5	26.6	27.7	28.7	29.7	30.6	31.6	32.5	34.2	35.9	37.4	38.9	40.4	46
48	16.5	18.1	19.6	21.0	22.3	23.6	24.8	26.0	27.1	28.2	29.2	30.2	31.2	32.2	33.1	34.9	36.6	38.2	39.7	41.2	48
50	16.8	18.4	19.9	21.4	22.7	24.0	25.2	26.4	27.6	28.7	29.8	30.8	31.8	32.8	33.7	35.5	37.2	38.9	40.5	42.0	50
52	17.1	18.7	20.2	21.7	23.1	24.4	25.7	26.9	28.0	29.2	30.3	31.3	32.3	33.3	34.3	36.2	37.9	39.6	41.2	42.8	52
54	17.3	19.0	20.6	22.0	23.5	24.8	26.1	27.3	28.5	29.7	30.8	31.8	32.9	33.9	34.9	36.8	38.6	40.3	41.9	43.5	54
56	17.6	19.3	20.9	22.4	23.8	25.2	26.5	27.7	28.9	30.1	31.2	32.3	33.4	34.4	35.4	37.4	39.2	41.0	42.7	44.3	56
58	17.8	19.5	21.2	22.7	24.2	25.5	26.9	28.2	29.4	30.6	31.7	32.8	33.9	35.0	36.0	38.0	39.8	41.6	43.3	45.0	58
60	18.1	19.8	21.5	23.0	24.5	25.9	27.3	28.6	29.8	31.0	32.2	33.3	34.4	35.5	36.5	38.5	40.4	42.3	44.0	45.7	60
62		20.1	21.7	23.3	24.8	26.3	27.6	28.9	30.2	31.5	32.6	33.8	34.9	36.0	37.1	39.1	41.0	42.9	44.7	46.4	62
64		20.3	22.0	23.6	25.1	26.6	28.0	29.3	30.6	31.9	33.1	34.3	35.4	36.5	37.6	39.6	41.6	43.5	45.3	47.1	64
66		20.6	22.3	23.9	25.5	26.9	28.4	29.7	31.0	32.3	33.5	34.7	35.9	37.0	38.1	40.2	42.2	44.1	46.0	47.7	66
68		20.8	22.6	24.2	25.8	27.3	28.7	30.1	31.4	32.7	33.9	35.2	36.3	37.5	38.6	40.7	42.8	44.7	46.6	48.4	68
70		21.1	22.6	24.5	26.1	27.6	29.1	30.4	31.8	33.1	34.4	35.6	36.8	37.9	39.1	41.2	43.3	45.3	47.2	49.0	70
72			23.1	24.8	26.4	27.9	29.4	30.8	32.2	33.5	34.8	36.0	37.2	38.4	39.5	41.7	43.8	45.8	47.8	49.6	72
74			23.3	25.1	26.7	28.2	29.7	31.2	32.5	33.9	35.2	36.4	37.7	38.8	40.0	42.2	44.4	46.4	48.4	50.3	74
76			23.6	25.3	27.0	28.5	30.0	31.5	32.9	34.3	35.6	36.8	38.1	39.3	40.5	42.7	44.9	47.0	48.9	50.9	76
78			23.8	25.6	27.3	28.8	30.4	31.8	33.3	34.6	36.0	37.2	38.5	39.7	40.9	43.2	45.4	47.5	49.5	51.4	78
80			24.1	25.8	27.5	29.1	30.7	32.2	33.6	35.0	36.3	37.6	38.9	40.2	41.4	43.7	45.9	48.0	50.1	52.0	80
82				26.1	27.8	29.4	31.0	32.5	34.0	35.4	36.7	38.0	39.3	40.6	41.8	44.1	46.4	48.5	50.6	52.6	82
84				26.4	28.1	29.7	31.3	32.8	34.3	35.7	37.1	38.4	39.7	41.0	42.2	44.6	46.9	49.0	51.1	53.2	84
86				26.6	28.3	30.0	31.6	33.1	34.6	36.1	37.4	38.8	40.1	41.4	42.6	45.0	47.3	49.6	51.7	53.7	86
88				26.9	28.6	30.3	31.9	33.4	34.9	36.4	37.8	39.2	40.5	41.8	43.1	45.5	47.8	50.0	52.2	54.3	88
90				27.1	28.9	30.6	32.2	33.8	35.3	36.7	38.2	39.5	40.9	42.2	43.5	45.9	48.3	50.5	52.7	54.8	90
92					29.1	30.8	32.5	34.1	35.6	37.1	38.5	39.9	41.3	42.6	43.9	46.4	48.7	51.0	53.2	55.3	92
96					29.6	31.4	33.0	34.7	36.2	37.7	39.2	40.6	42.0	43.3	44.7	47.2	49.6	52.0	54.2	56.4	96

Cont. Table 14-2 CIRCULAR EQUIVALENTS OF RECTANGULAR DUCTS FOR EQUAL FRICTION AND CAPACITY (U.S. Units) (2)

Dimensions in Inches

Side Rectangular Duct	32	34	36	38	40	42	44	46	48	50	52	56	60	64	68	72	76	80	84	88	Side Rectangular Duct
32	35.0																				32
34	36.1	37.2																			34
36	37.1	38.2	39.4																		36
38	38.1	39.3	40.4	41.5																	38
40	39.0	40.3	41.5	42.6	43.7																40
42	40.0	41.3	42.5	43.7	44.8	45.9															42
44	40.9	42.2	43.5	44.7	45.8	47.0	48.1														44
46	41.8	43.1	44.4	45.7	46.9	48.0	49.2	50.3													46
48	42.6	44.0	45.3	46.6	47.9	49.1	50.2	51.4	52.5												48
50	43.6	44.9	46.2	47.5	48.8	50.0	51.2	52.4	53.6	54.7											50
52	44.3	45.7	47.1	48.4	49.7	51.0	52.2	53.4	54.6	55.7	56.8										52
54	45.1	46.5	48.0	49.3	50.7	52.0	53.2	54.4	55.6	56.8	57.9										54
56	45.8	47.3	48.8	50.2	51.6	52.9	54.2	55.4	56.6	57.8	59.0	61.2									56
58	46.6	48.1	49.6	51.0	52.4	53.8	55.1	56.4	57.6	58.8	60.0	62.3									58
60	47.3	48.9	50.4	51.9	53.3	54.7	60.0	57.3	58.6	59.8	61.0	63.4	65.6								60
62	48.0	49.6	51.2	52.7	54.1	55.5	56.9	58.2	59.5	60.8	62.0	64.4	66.7								62
64	48.7	50.4	51.9	53.5	54.9	56.4	57.8	59.1	60.4	61.7	63.0	65.4	67.7	70.0							64
66	49.4	51.1	52.7	54.2	55.7	57.2	58.6	60.0	61.3	62.6	63.9	66.4	68.8	71.0							66
68	50.1	51.8	53.4	55.0	56.5	58.0	59.4	60.8	62.2	63.6	64.9	67.4	69.8	72.1	74.3						68
70	50.8	52.5	54.1	55.7	57.3	58.8	60.3	61.7	63.1	64.4	65.8	68.3	70.8	73.2	75.4						70
72	51.4	53.2	54.8	56.5	58.0	59.6	61.1	62.5	63.9	65.3	66.7	69.3	71.8	74.2	76.5	78.7					72
74	52.1	53.8	55.5	57.2	58.8	60.3	61.9	63.3	64.8	66.2	67.5	70.2	72.7	75.2	77.5	79.8					74
76	52.7	54.5	56.2	57.9	59.5	61.1	62.6	64.1	65.6	67.0	68.4	71.1	73.7	76.2	78.6	80.9	83.1				76
78	53.3	55.1	56.9	58.6	60.2	61.8	63.4	64.9	66.4	67.9	69.3	72.0	74.6	77.1	79.6	81.9	84.2				78
80	53.9	55.8	57.5	59.3	60.9	62.6	64.1	65.7	67.2	68.7	70.1	72.9	75.4	78.1	80.6	82.9	85.2	87.5			80
82	54.5	56.4	58.2	59.9	61.6	63.3	64.9	66.5	68.0	69.5	70.9	73.7	76.4	79.0	81.5	84.0	86.3	88.5			82
84	55.1	57.0	58.8	60.6	62.3	64.0	65.6	67.2	68.7	70.3	71.7	74.6	77.3	80.0	82.5	85.0	87.3	89.6	91.8		84
86	55.7	57.6	59.4	61.2	63.0	64.7	66.3	67.9	69.5	71.0	72.5	75.4	78.2	80.9	83.5	85.9	88.3	90.7	92.9		86
88	56.3	58.2	60.1	61.9	63.6	65.4	67.0	68.7	70.2	71.8	73.3	76.3	79.1	81.8	84.4	86.9	89.3	91.7	94.0	96.2	88
90	56.8	58.8	60.7	62.5	64.3	66.0	67.7	69.4	71.0	72.6	74.1	77.1	79.9	82.7	85.3	87.9	90.3	92.7	95.0	97.3	90
92	57.4	59.3	61.3	63.1	64.9	66.7	68.4	70.1	71.7	73.3	74.9	77.9	80.8	83.5	86.2	88.8	91.3	93.7	96.1	98.4	92
94	57.9	59.9	61.9	63.7	65.6	67.3	69.1	70.8	72.4	74.0	75.6	78.7	81.6	84.4	87.1	89.7	92.3	94.7	97.1	99.4	94
96	58.4	60.5	62.4	64.3	66.2	68.0	69.7	71.5	73.1	74.8	76.3	79.4	82.4	85.3	88.0	90.7	93.2	95.7	98.1	100.5	96

Equation for Circular Equivalent of a Rectangular Duct:

$$D_e = 1.30\,[(ab)^{0.625}/(a + b)^{0.250}]$$

where

a = length of one side of rectangular duct, inches.
b = length of adjacent side of rectangular duct, inches.
D_e = circular equivalent of rectangular duct for equal friction and capacity, inches.

Table 14-3 CIRCULAR EQUIVALENTS OF RECTANGULAR DUCTS FOR EQUAL FRICTION AND CAPACITY (Metric Units) (2)

Dimensions in mm

Side Rectangular Duct	100	125	150	175	200	225	250	275	300	350	400	450	500	550	600	650	700	750	800	900	Side Rectangular Duct
100	109																				100
125	122	137																			125
150	133	150	164																		150
175	143	161	177	191																	175
200	152	172	189	204	219																200
225	161	181	200	216	232	246															225
250	169	190	210	228	244	259	273														250
275	176	199	220	238	256	272	287	301													275
300	183	207	229	248	266	283	299	314	328												300
350	195	222	245	267	286	305	322	339	354	383											350
400	207	235	260	283	305	325	343	361	378	409	437										400
450	217	247	274	299	321	343	363	382	400	433	464	492									450
500	227	258	287	313	337	360	381	401	420	455	488	518	547								500
550	236	269	299	326	352	375	398	419	439	477	511	543	573	601							550
600	245	279	310	339	365	390	414	436	457	496	533	567	598	628	656						600
650	253	289	321	351	378	404	429	452	474	515	553	589	622	653	683	711					650
700	261	298	331	362	391	418	443	467	490	533	573	610	644	677	708	737	765				700
750	268	306	341	373	402	430	457	482	506	550	592	630	666	700	732	763	792	820			750
800	275	314	350	383	414	442	470	496	520	567	609	649	687	722	755	787	818	847	875		800
900	289	330	367	402	435	465	494	522	548	597	643	686	726	763	799	833	866	897	927	984	900
1000	301	344	384	420	454	486	517	546	574	626	674	719	762	802	840	876	911	944	976	1037	1000
1100	313	358	399	437	473	506	538	569	598	652	703	751	795	838	878	916	953	988	1022	1086	1100
1200	324	370	413	453	490	525	558	590	620	677	731	780	827	872	914	954	993	1030	1066	1133	1200
1300	334	382	426	468	506	543	577	610	642	701	757	808	857	904	948	990	1031	1069	1107	1177	1300
1400	344	394	439	482	522	559	595	629	662	724	781	835	886	934	980	1024	1066	1107	1146	1220	1400
1500	353	404	452	495	536	575	612	648	681	745	805	860	913	963	1011	1057	1100	1143	1183	1260	1500
1600	362	415	463	508	551	591	629	665	700	766	827	885	939	991	1041	1088	1133	1177	1219	1298	1600
1700	371	425	475	521	564	605	644	682	718	785	849	908	964	1018	1069	1118	1164	1209	1253	1335	1700
1800	379	434	485	533	577	619	660	698	735	804	869	930	988	1043	1096	1146	1195	1241	1286	1371	1800
1900	387	444	496	544	590	633	674	713	751	823	889	952	1012	1068	1122	1174	1224	1271	1318	1405	1900
2000	395	453	506	555	602	646	688	728	767	840	908	973	1034	1092	1147	1200	1252	1301	1348	1438	2000
2100	402	461	516	566	614	659	702	743	782	857	927	993	1055	1115	1172	1226	1279	1329	1378	1470	2100
2200	410	470	525	577	625	671	715	757	797	874	945	1013	1076	1137	1195	1251	1305	1356	1406	1501	2200
2300	417	478	534	587	636	683	728	771	812	890	963	1031	1097	1159	1218	1275	1330	1383	1434	1532	2300
2400	424	486	543	597	647	695	740	784	826	905	980	1050	1116	1180	1241	1299	1355	1409	1461	1561	2400
2500	430	494	552	606	658	706	753	797	840	920	996	1068	1136	1200	1262	1322	1379	1434	1488	1589	2500
2600	437	501	560	616	668	717	764	810	853	935	1012	1085	1154	1220	1283	1344	1402	1459	1513	1617	2600
2700	443	509	569	625	678	728	776	822	866	950	1028	1102	1173	1240	1304	1366	1425	1483	1538	1644	2700
2800	450	516	577	634	688	738	787	834	879	964	1043	1119	1190	1259	1324	1387	1447	1506	1562	1670	2800
2900	456	523	585	643	697	749	798	845	891	977	1058	1135	1208	1277	1344	1408	1469	1529	1586	1696	2900
Side Rectangular Duct	100	125	150	175	200	225	250	275	300	350	400	450	500	550	600	650	700	750	800	900	Side Rectangular Duct

Cont. Table 14-3 CIRCULAR EQUIVALENTS OF RECTANGULAR DUCTS FOR EQUAL FRICTION AND CAPACITY (Metric Units) (2)

Dimensions in mm

Side Rectangular Duct	1000	1100	1200	1300	1400	1500	1600	1700	1800	1900	2000	2100	2200	2300	2400	2500	2600	2700	2800	2900	Side Rectangular Duct
1000	1093																				1000
1100	1146	1202																			1100
1200	1196	1256	1312																		1200
1300	1244	1306	1365	1421																	1300
1400	1289	1354	1416	1475	1530																1400
1500	1332	1400	1464	1526	1584	1640															1500
1600	1373	1444	1511	1574	1635	1693	1749														1600
1700	1413	1486	1555	1621	1684	1745	1803	1858													1700
1800	1451	1527	1598	1667	1732	1794	1854	1912	1968												1800
1900	1488	1566	1640	1710	1778	1842	1904	1964	2021	2077											1900
2000	1523	1604	1680	1753	1822	1889	1952	2014	2073	2131	2186										2000
2100	1558	1640	1719	1793	1865	1933	1999	2063	2124	2183	2240	2296									2100
2200	1591	1676	1756	1833	1906	1977	2044	2110	2173	2233	2292	2350	2405								2200
2300	1623	1710	1793	1871	1947	2019	2088	2155	2220	2283	2343	2402	2459	2514							2300
2400	1655	1744	1828	1909	1986	2060	2131	2200	2266	2330	2393	2453	2511	2568	2624						2400
2500	1685	1776	1862	1945	2024	2100	2173	2243	2311	2377	2441	2502	2562	2621	2678	2733					2500
2600	1715	1808	1896	1980	2061	2139	2213	2285	2355	2422	2487	2551	2612	2672	2730	2787	2842				2600
2700	1744	1839	1929	2015	2097	2177	2253	2327	2398	2466	2533	2598	2661	2722	2782	2840	2896	2952			2700
2800	1772	1869	1961	2048	2133	2214	2292	2367	2439	2510	2578	2644	2708	2771	2832	2891	2949	3006	3061		2800
2900	1800	1898	1992	2081	2167	2250	2329	2406	2480	2552	2621	2689	2755	2819	2881	2941	3001	3058	3115	3170	2900
Side Rectangular Duct	1000	1100	1200	1300	1400	1500	1600	1700	1800	1900	2000	2100	2200	2300	2400	2500	2600	2700	2800	2900	Side Rectangular Duct

Equation for Circular Equivalent of a Rectangular Duct:

$$D_e = 1.30\,[(ab)^{0.625}/(a + b)^{0.250}]$$

where

a = length of one side of rectangular duct, mm.
b = length of adjacent side of rectangular duct, mm.
D_e = circular equivalent of rectangular duct for equal friction and capacity, mm.

Table 14-4 SPIRAL FLAT-OVAL DUCT (Nominal Sizes—U.S. Units)
(Diameter of the round duct which will have the capacity and friction equivalent to the actual duct size.)

	3	4	5	6	7	8	9	10	11	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
7		5.7																						
8	5.1		6.6	6.9																				
9	5.6	6.2		7.7																				
10		6.7	7.3		8.7	9.0																		
11	6.0		7.9	8.4		9.8																		
12	6.4	7.2		8.9	9.4		10.8	11.0																
13		7.6	8.4		10.1	10.6		11.9																
14	6.7		8.8	9.6		11.2	11.5																	
15	7.0	8.0		10.1	10.7					13.8														
16			9.3					13.4	13.6															
17	7.3	8.4					12.9																	
18					11.7	12.4				15.3														
19			10.0	11.0					15.0															
20							14.0	14.7			17.5													
21					12.6	13.5				16.7			19.9											
22				11.8					16.3			19.5												
23							15.1	15.7			18.9													
24						14.4							21.6											
25				12.5						18.0		20.9			23.9									
26								16.7						23.6										
27						15.2					20.2		23.1			25.9								
28				13.2						19.1					25.6									
29								17.7				22.3		25.2			27.9							
30						15.9					21.3					28.1								
31				13.8						20.1			24.5		27.2			29.9						
32								18.5				23.5					29.7							
33						16.6					22.4			26.6		29.3			32.0					
34				14.3						20.9			25.7					31.7						
35								19.3				24.7			28.7		31.3			34.0				
36						17.3					23.4			27.9					33.7					
37				14.9						21.9			27.0			30.8		33.4			36.0			
38								20.1				25.7			30.0					35.8				
39						17.9					24.4			29.2			32.8		35.4			38.0		
40										22.7			28.1			32.2					37.8			
41				15.4				20.8				26.8			31.3			34.9		37.4			40.0	
42											25.3			30.3			34.3					39.8		
43						18.6				23.5			29.1			33.5			37.0		39.5			42.0
44				15.9								27.7			32.5			36.4					41.8	
45								21.5			26.1			31.4			35.6			39.0		41.5		
46						19.1							30.2			34.7			38.5					43.8
47										24.3		28.6			33.7			37.8			41.1		43.5	
48								22.1						32.5			36.9			40.5				
49						19.6					26.9		31.1			35.9			39.9			43.1		45.6
50										25.0					34.8			39.1			42.6			
51								22.8				29.4		33.4			38.1			42.0			45.2	
52						20.2					27.7					37.0			41.2			44.7		
53										25.7			32.0		35.8			40.3			44.1			47.2
54								23.3				30.2					39.3			43.3			46.7	
55											28.4			34.4		38.1			42.5			46.2		
56										26.3			32.9					41.5			45.5			48.8
57								23.8				31.0			36.7		40.4			44.6			48.2	
58											29.1			35.3					43.7			47.6		
59										26.9			33.7			39.2		42.6			46.8			50.4
60								24.4			29.6	31.8		36.1	37.8		41.3		44.5	45.9		48.5	49.7	
62										27.5			34.5			40.1		43.5			48.1			51.8
64								25.1			30.5	32.7		37.1	38.9		42.5		46.0			50.2		
66								25.4		28.3		33.3	35.4		39.6			44.8		48.0	49.7		52.4	
68										28.6	31.2			38.0		42.0			47.1			51.9		54.5
70								26.0			31.5	34.1	36.2		40.6		44.4	46.0		49.4	50.9		54.1	
72								26.3		29.2									48.6		51.6	53.1		56.2

Table 14-5 SPIRAL FLAT-OVAL DUCT (Nominal Sizes—Metric Units)
(Diameter of round duct which will have the capacity and friction equivalent to the actual duct size.)

	75	100	125	150	175	200	225	250	275	300	350	400	450	500	550	600	650	700	750	800	850	900	950	1000
175		145																						
200	130		168	175																				
225	142	157		196																				
250		170	185		221	229																		
275	152		201	213		249																		
300	163	183		226	239		274	279																
325		193	213		257	269		302																
350	170		224	244		284	292																	
375	178	203		257	272					351														
400			236					340	345															
425	185	213					328																	
450					297	315				389														
475			254	279					381															
500							356	373			445													
525					320	342				424			505											
550				300					414			495												
575							384	399			480													
600						366							549											
625				318						457		531			607									
650								424						599										
675						386					513		587			658								
700				335						485					650									
725								450				566		640			709							
750						404					541					714								
775				351						511			622		691			759						
800								470				597					754							
825						422					569		676		744				813					
850				363						531			653					805						
875								490				627			729		795			864				
900						439					594			709					856					
925				378						556			686			782		848			914			
950								511				653			762					909				
975						455					620			742			833		899			965		
1000										577			714			818					960			
1025				391				528				681			795			886		950			1016	
1050											643			770			871					1011		
1075						472				597			739			851			939		1004			1067
1100				404								504			826			925					1062	
1125								546			663			798			904			991		1054		
1150						485							767			881			978					1113
1175										617		726			856			960			1044		1105	
1200								561						826			937			1029				
1225						498					683		790			912			1013			1095		1158
1250										635					884			993			1082			
1275								579				747		848			968			1067			1148	
1300						513					704					940			1046			1135		
1325										653			813		909			1023			1120			1199
1350								592				767					998			1100			1186	
1375											721			874		968			1080			1174		
1400										668			836					1054			1156			1240
1425								605				787			932		1026			1133			1224	
1450											739			897					1110			1209		
1475										683			856			996		1082			1189			1280
1500								620			752	808		917	960		1049		1130	1166		1232	1262	
1550										699			876			1018		1105			1222			1316
1600								638			775	830		942	988		1079		1168			1275		
1650								645		719		846	899		1006			1138		1219	1262		1331	
1700										726	792			965		1067			1196			1318		1384
1750								660			800	866	919		1031		1128	1168		1255	1293		1374	
1800								668		742									1234		1311	1379		1427

Table 14-6 VELOCITIES/VELOCITY PRESSURES (U.S. Units)

Velocity fpm	Velocity Pressure in. wg.	Velocity fpm	Velocity Pressure in. wg.	Velocity fpm	Velocity Pressure in. wg.	Velocity fpm	Velocity Pressure in. wg.	Velocity fpm	Velocity Pressure in. wg.
300	0.01	2050	0.26	3800	0.90	5550	1.92	7300	3.32
350	0.01	2100	0.27	3850	0.92	5600	1.95	7350	3.37
400	0.01	2150	0.29	3900	0.95	5650	1.99	7400	3.41
450	0.01	2200	0.30	3950	0.97	5700	2.02	7450	3.46
500	0.02	2250	0.32	4000	1.00	5750	2.06	7500	3.51
550	0.02	2300	0.33	4050	1.02	5800	2.10	7550	3.55
600	0.02	2350	0.34	4100	1.05	5850	2.13	7600	3.60
650	0.03	2400	0.36	4150	1.07	5900	2.17	7650	3.65
700	0.03	2450	0.37	4200	1.10	5950	2.21	7700	3.70
750	0.04	2500	0.39	4250	1.13	6000	2.24	7750	3.74
800	0.04	2550	0.41	4300	1.15	6050	2.28	7800	3.79
850	0.05	2600	0.42	4350	1.18	6100	2.32	7850	3.84
900	0.05	2650	0.44	4400	1.21	6150	2.36	7900	3.89
950	0.06	2700	0.45	4450	1.23	6200	2.40	7950	3.94
1000	0.06	2750	0.47	4500	1.26	6250	2.43	8000	3.99
1050	0.07	2800	0.49	4550	1.29	6300	2.47	8050	4.04
1100	0.08	2850	0.51	4600	1.32	6350	2.51	8100	4.09
1150	0.08	2900	0.52	4650	1.35	6400	2.55	8150	4.14
1200	0.09	2950	0.54	4700	1.38	6450	2.59	8200	4.19
1250	0.10	3000	0.56	4750	1.41	6500	2.63	8250	4.24
1300	0.11	3050	0.58	4800	1.44	6550	2.67	8300	4.29
1350	0.11	3100	0.60	4850	1.47	6600	2.71	8350	4.35
1400	0.12	3150	0.62	4900	1.50	6650	2.76	8400	4.40
1450	0.13	3200	0.64	4950	1.53	6700	2.80	8450	4.45
1500	0.14	3250	0.66	5000	1.56	6750	2.84	8500	4.50
1550	0.15	3300	0.68	5050	1.59	6800	2.88	8550	4.56
1600	0.16	3350	0.70	5100	1.62	6850	2.92	8600	4.61
1650	0.17	3400	0.72	5150	1.65	6900	2.97	8650	4.66
1700	0.18	3450	0.74	5200	1.69	6950	3.01	8700	4.72
1750	0.19	3500	0.76	5250	1.72	7000	3.05	8750	4.77
1800	0.20	3550	0.79	5300	1.75	7050	3.10	8800	4.83
1850	0.21	3600	0.81	5350	1.78	7100	3.14	8850	4.88
1900	0.22	3650	0.83	5400	1.82	7150	3.19	8900	4.94
1950	0.24	3700	0.85	5450	1.85	7200	3.23	8950	4.99
2000	0.25	3750	0.88	5500	1.89	7250	3.28	9000	5.05

$$\text{Velocity} = 4005\sqrt{V_p} \text{ (or) } V_p = \left(\frac{\text{Velocity}}{4005}\right)^2$$

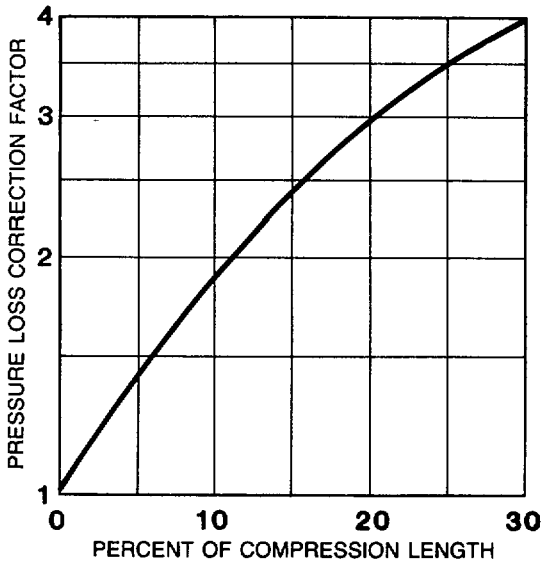


Figure 14-4 CORRECTION FACTOR FOR UNEXTENDED FLEXIBLE DUCT (2)

Table 14-7 VELOCITIES/VELOCITY PRESSURES (Metric Units)

Velocity (m/s)	Velocity Pressure (Pa)	Velocity (m/s)	Velocity Pressure (Pa)	Velocity (m/s)	Velocity Pressure (Pa)	Velocity (m/s)	Velocity Pressure (Pa)	Velocity (m/s)	Velocity Pressure (Pa)
1.0	0.6	10.0	60	19.0	217	28.0	472	37.0	824
1.2	0.9	10.2	63	19.2	222	28.2	479	37.2	833
1.4	1.2	10.4	65	19.4	227	28.4	486	37.4	842
1.6	1.5	10.6	68	19.6	231	28.6	493	37.6	851
1.8	2.0	10.8	70	19.8	236	28.8	499	37.8	860
2.0	2.4	11.0	73	20.0	241	29.0	506	38.0	870
2.2	2.9	11.2	76	20.2	246	29.2	513	38.2	879
2.4	3.5	11.4	78	20.4	251	29.4	521	38.4	888
2.6	4.1	11.6	81	20.6	256	29.6	528	38.6	897
2.8	4.7	11.8	84	20.8	261	29.8	535	38.8	907
3.0	5.4	12.0	87	21.0	266	30.0	542	39.0	916
3.2	6.2	12.2	90	21.2	271	30.2	549	39.2	925
3.4	7.0	12.4	93	21.4	276	30.4	557	39.4	935
3.6	7.8	12.6	96	21.6	281	30.6	564	39.6	944
3.8	8.7	12.8	99	21.8	286	30.8	571	39.8	954
4.0	9.6	13.0	102	22.0	291	31.0	579	40.0	963
4.2	10.6	13.2	105	22.2	297	31.2	586	40.2	973
4.4	11.7	13.4	108	22.4	302	31.4	594	40.4	983
4.6	12.7	13.6	111	22.6	308	31.6	601	40.6	993
4.8	13.9	13.8	115	22.8	313	31.8	609	40.8	1002
5.0	15.1	14.0	118	23.0	319	32.0	617	41.0	1012
5.2	16.3	14.2	121	23.2	324	32.2	624	41.2	1022
5.4	17.6	14.4	125	23.4	330	32.4	632	41.4	1032
5.6	18.9	14.6	128	23.6	335	32.6	640	41.6	1042
5.8	20.3	14.8	132	23.8	341	32.8	648	41.8	1052
6.0	21.7	15.0	135	24.0	347	33.0	656	42.0	1062
6.2	23.1	15.2	139	24.2	353	33.2	664	42.2	1072
6.4	24.7	15.4	143	24.4	359	33.4	672	42.4	1083
6.6	26.2	15.6	147	24.6	364	33.6	680	42.6	1093
6.8	27.8	15.8	150	24.8	370	33.8	688	42.8	1103
7.0	29.5	16.0	154	25.0	376	34.0	696	43.0	1113
7.2	31.2	16.2	158	25.2	382	34.2	704	43.2	1124
7.4	33.0	16.4	162	25.4	389	34.4	713	43.4	1134
7.6	34.8	16.6	166	25.6	395	34.6	721	43.6	1145
7.8	36.6	16.8	170	25.8	401	34.8	729	43.8	1155
8.0	38.5	17.0	174	26.0	407	35.0	738	44.0	1166
8.2	40.5	17.2	178	26.2	413	35.2	746	44.2	1176
8.4	42.5	17.4	182	26.4	420	35.4	755	44.4	1187
8.6	44.5	17.6	187	26.6	426	35.6	763	44.6	1198
8.8	46.6	17.8	191	26.8	433	35.8	772	44.8	1209
9.0	48.8	18.0	195	27.0	439	36.0	780	45.0	1219
9.2	51.0	18.2	199	27.2	446	36.2	789	45.2	1230
9.4	53.2	18.4	204	27.4	452	36.4	798	45.4	1241
9.6	55.5	18.6	208	27.6	459	36.6	807	45.6	1252
9.8	57.8	18.8	213	27.8	465	36.8	815	45.8	1263

Table 14-8 ANGULAR CONVERSION

Degrees	Radians
10°	0.175
20°	0.349
30°	0.524
40°	0.698
50°	0.873
60°	1.05
70°	1.22
80°	1.40
90°	1.57 ($\pi/2$)
135°	2.36
180°	3.14 (π)
360°	6.28 (2π)

Table 14-9 LOSS COEFFICIENTS FOR STRAIGHT-THROUGH FLOW

$\frac{V_s}{V_c}$	C	
	75% Regain	90% Regain
0.95	0.03	0.01
.91	.04	.02
.87	.06	.02
.83	.08	.03
.80	.09	.04
0.77	0.10	0.04
.74	.11	.04
.71	.12	.05
.69	.13	.05
.67	.14	.06
0.65	0.15	0.06
.63	.15	.06
.61	.16	.06
.59	.16	.07
.57	.17	.07
0.56	0.18	0.07
.54	.18	.07
.53	.18	.07
.51	.18	.07
.50	.19	.08

V_s = Downstream (System) Velocity
V_c = Upstream (System) Velocity

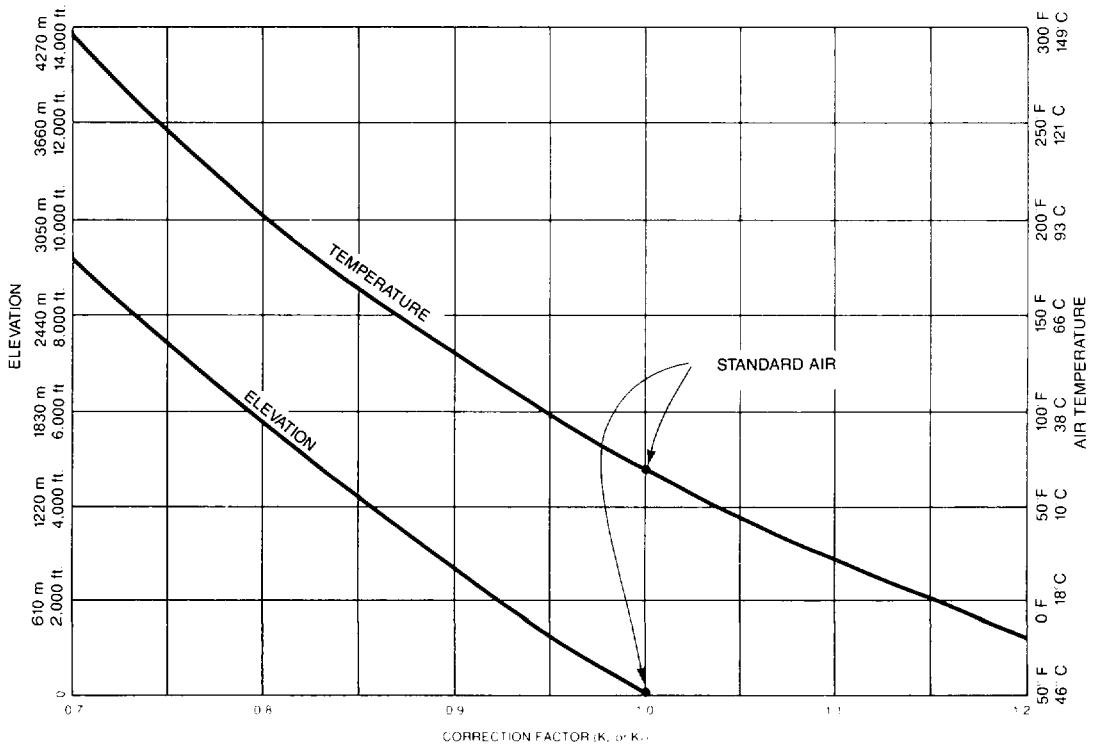


Figure 14-5 AIR DENSITY FRICTION CHART CORRECTION FACTORS

When an air distribution system is designed to operate above 2000 feet (610 m) altitude, below 32°F (0°C), or above 120°F (49°C) temperature, the duct friction loss obtained must be corrected for the air density. The actual airflow (cfm or l/s) is used to find the duct friction loss which is multiplied by the correction factor or factors from the above chart to obtain the actual friction loss.

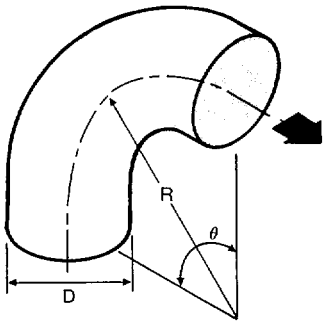
B FITTING LOSS COEFFICIENT TABLES

Duct Cross Section to which Coefficient "C" is referenced is at the top of each table. Negative numbers indicate that the static regain exceeds the dynamic pressure loss of the fitting.

Table 14-10 LOSS COEFFICIENTS, ELBOWS

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

A. Elbow, Smooth Radius (Die Stamped), Round (2)



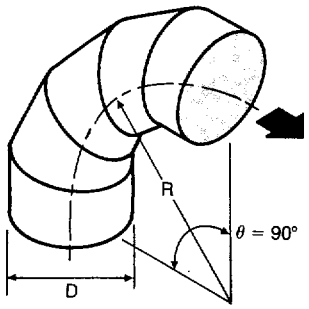
Coefficients for 90° Elbows: (See Note 1)

R/D	0.5	0.75	1.0	1.5	2.0	2.5
C	0.71	0.33	0.22	0.15	0.13	0.12

Note 1: For angles other than 90° multiply by the following factors:

θ	0°	20°	30°	45°	60°	75°	90°	110°	130°	150°	180°
K	0	0.31	0.45	0.60	0.78	0.90	1.00	1.13	1.20	1.28	1.40

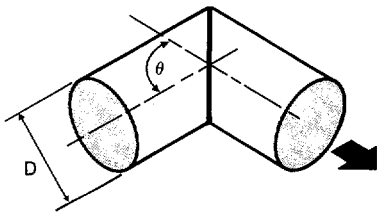
B. Elbow, Round, 3 to 5 pc — 90° (2)



Coefficient C

No. of Pieces	R/D 0.5	0.75	1.0	1.5	2.0
5	—	0.46	0.33	0.24	0.19
4	—	0.50	0.37	0.27	0.24
3	0.98	0.54	0.42	0.34	0.33

C. Elbow, Round, Mitered (15)



Coefficient C (See Note 2)

θ	20°	30°	45°	60°	75°	90°
C	0.08	0.16	0.34	0.55	0.81	1.2

Note 2: Correction factor for Reynolds number — K_{Re}

R_e 10^{-4}	1	2	3	4	6	8	10	≥14
K_{Re}	1.40	1.26	1.19	1.14	1.09	1.06	1.04	1.0

U.S. Units
For Standard Air:
$R_e = 8.56\ DV$

(Metric Units)
($R_e = 66.4\ DV$)

where:
D = duct diameter, inches (mm)
V = duct velocity, fpm (m/s)

For Rectangular Ducts:

$$D = \frac{2HW}{H + W}$$

Reynolds Number Correction Factor Chart

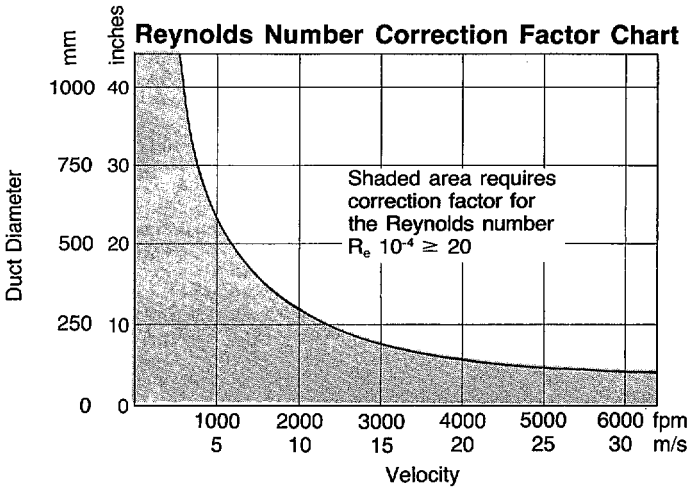
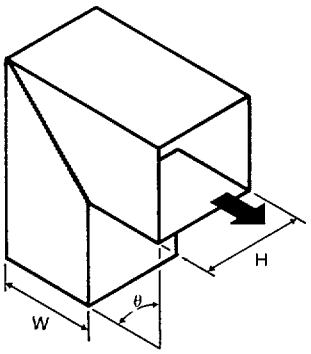


Table 14-10 LOSS COEFFICIENTS, ELBOWS (Cont.)

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

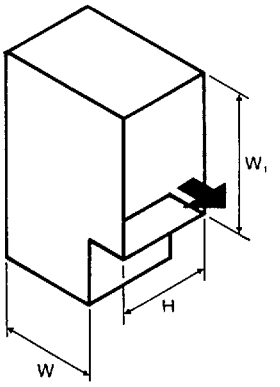
D. Elbow, Rectangular, Mitered (15)



Coefficient C (See Note 2—Page 14.19)

θ	H/W										
	0.25	0.5	0.75	1.0	1.5	2.0	3.0	4.0	5.0	6.0	8.0
20°	0.08	0.08	0.08	0.07	0.07	0.07	0.06	0.06	0.05	0.05	0.05
30°	0.18	0.17	0.17	0.16	0.15	0.15	0.13	0.13	0.12	0.12	0.11
45°	0.38	0.37	0.36	0.34	0.33	0.31	0.28	0.27	0.26	0.25	0.24
60°	0.60	0.59	0.57	0.55	0.52	0.49	0.46	0.43	0.41	0.39	0.38
75°	0.89	0.87	0.84	0.81	0.77	0.73	0.67	0.63	0.61	0.58	0.57
90°	1.3	1.3	1.2	1.2	1.1	1.1	0.98	0.92	0.89	0.85	0.83

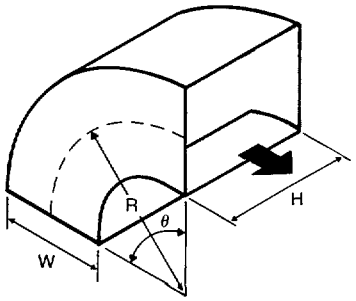
E. Elbow, Rectangular, Mitered with Converging or Diverging Flow (15)



Coefficient C (See Note 2—Page 14.19)

H/W	W_1/W					
	0.6	0.8	1.2	1.4	1.6	2.0
0.25	1.8	1.4	1.1	1.1	1.1	1.1
1.0	1.7	1.4	1.0	0.95	0.90	0.84
4.0	1.5	1.1	0.81	0.76	0.72	0.66
∞	1.5	1.0	0.69	0.63	0.60	0.55

F. Elbow, Rectangular, Smooth Radius without Vanes (15)



Coefficients for 90° elbows: (See Note 1)

Coefficient C (See Note 3)

R/W	H/W										
	0.25	0.5	0.75	1.0	1.5	2.0	3.0	4.0	5.0	6.0	8.0
0.5	1.5	1.4	1.3	1.2	1.1	1.0	1.0	1.1	1.1	1.2	1.2
0.75	0.57	0.52	0.48	0.44	0.40	0.39	0.39	0.40	0.42	0.43	0.44
1.0	0.27	0.25	0.23	0.21	0.19	0.18	0.18	0.19	0.20	0.27	0.21
1.5	0.22	0.20	0.19	0.17	0.15	0.14	0.14	0.15	0.16	0.17	0.17
2.0	0.20	0.18	0.16	0.15	0.14	0.13	0.13	0.14	0.14	0.15	0.15

Note 3: Correction Factor for Reynolds number — K_{Re}

R/W	R_e 10^{-4}								
	1	2	3	4	6	8	10	14	≥20
0.5	1.40	1.26	1.19	1.14	1.09	1.06	1.04	1.0	1.0
≥0.75	2.0	1.77	1.64	1.56	1.46	1.38	1.30	1.15	1.0

U.S. Units
For Standard Air:
R_e = 8.56 DV

(Metric Units)
(R_e = 66.4 DV)

where:
D = duct diameter, inches (mm)
V = duct velocity, fpm (m/s)

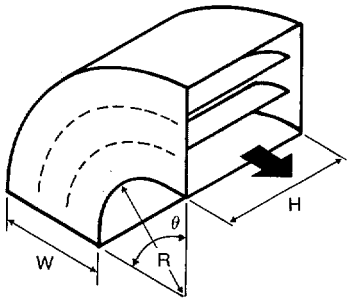
For Rectangular Ducts:

$$D = \frac{2HW}{H + W}$$

Table 14-10 LOSS COEFFICIENTS, ELBOWS (Cont.)

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

G. Elbow, Rectangular, Smooth Radius with Splitter Vanes (2)



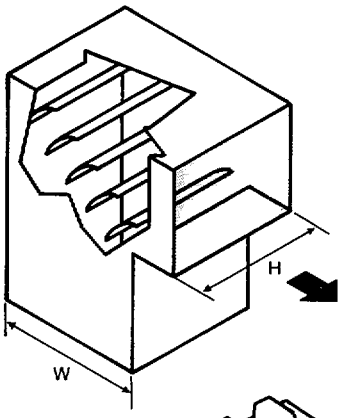
NOTES FOR THIS FIGURE ONLY:
A) See Page 5.14 to calculate splitter vane spacing.
B) CR = Curve Ratio
C) Use correction factors in Note 1 on Page 14.19 for elbows other than 90°.

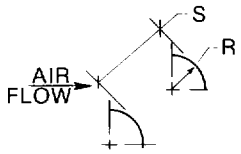
	R/W	CR	Coefficient C, H/W 0.25	0.5	1.0	1.5	2.0	3.0	4.0	5.0	6.0	7.0	8.0
1 splitter vane	0.05	0.218	0.52	0.40	0.43	0.49	0.55	0.66	0.75	0.84	0.93	1.0	1.1
	0.10	0.302	0.36	0.27	0.25	0.28	0.30	0.35	0.39	0.42	0.46	0.49	0.52
	0.15	0.361	0.28	0.21	0.18	0.19	0.20	0.22	0.25	0.26	0.28	0.30	0.32
	0.20	0.408	0.22	0.16	0.14	0.14	0.15	0.16	0.17	0.18	0.19	0.20	0.21
	0.25	0.447	0.18	0.13	0.11	0.11	0.11	0.12	0.13	0.14	0.14	0.15	0.15
	0.30	0.480	0.15	0.11	0.09	0.09	0.09	0.09	0.10	0.10	0.11	0.11	0.12
	0.35	0.509	0.13	0.09	0.08	0.07	0.07	0.08	0.08	0.08	0.08	0.09	0.09
	0.40	0.535	0.11	0.08	0.07	0.06	0.06	0.06	0.06	0.07	0.07	0.07	0.07
	0.45	0.557	0.10	0.07	0.06	0.05	0.05	0.05	0.05	0.05	0.06	0.06	0.06
	0.50	0.577	0.09	0.06	0.05	0.05	0.04	0.04	0.04	0.05	0.05	0.05	0.05
2 splitter vanes	0.05	0.362	0.26	0.20	0.22	0.25	0.28	0.33	0.37	0.41	0.45	0.48	0.51
	0.10	0.450	0.17	0.13	0.11	0.12	0.13	0.15	0.16	0.17	0.19	0.20	0.21
	0.15	0.507	0.12	0.09	0.08	0.08	0.08	0.09	0.10	0.10	0.11	0.11	0.11
	0.20	0.550	0.09	0.07	0.06	0.05	0.06	0.06	0.06	0.06	0.07	0.07	0.07
	0.25	0.585	0.08	0.05	0.04	0.04	0.04	0.04	0.05	0.05	0.05	0.05	0.05
	0.30	0.613	0.06	0.04	0.03	0.03	0.03	0.03	0.03	0.03	0.04	0.04	0.04
	0.35	0.638	0.05	0.04	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03
	0.40	0.659	0.05	0.03	0.03	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
	0.45	0.677	0.04	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
	0.50	0.693	0.03	0.02	0.02	0.02	0.02	0.01	0.01	0.01	0.01	0.01	0.01
3 splitter vanes	0.05	0.467	0.11	0.10	0.12	0.13	0.14	0.16	0.18	0.19	0.21	0.22	0.23
	0.10	0.549	0.07	0.05	0.06	0.06	0.06	0.07	0.07	0.08	0.08	0.08	0.09
	0.15	0.601	0.05	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.05	0.05
	0.20	0.639	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03
	0.25	0.669	0.03	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
	0.30	0.693	0.03	0.02	0.02	0.02	0.02	0.01	0.01	0.01	0.01	0.01	0.01
	0.35	0.714	0.02	0.02	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
	0.40	0.731	0.02	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
	0.45	0.746	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
	0.50	0.760	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01

Table 14-10 LOSS COEFFICIENTS, ELBOWS (Cont.)

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

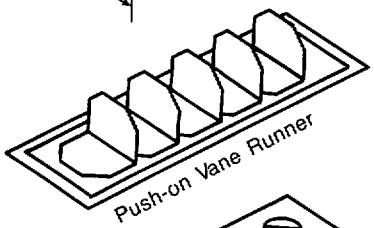
H. Elbow, Rectangular, Mitered with Turning Vanes
(See Chapter 5, Sections E and H for additional information and data.)

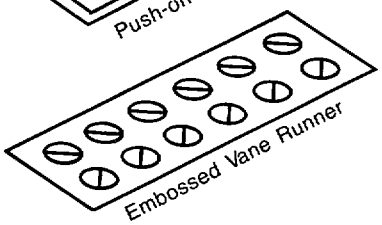


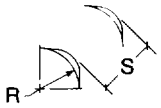


Loss Coefficients (C) for Single Thickness Vanes

Dimensions, Inches (mm)		Velocity, fpm (m/s)			
R	S	1000 (5)	1500 (7.5)	2000 (10)	2500 (12.5)
2.0 (50)	1.5 (38)	0.24	0.23	0.22	0.20
4.5 (114)	3.25 (83)	0.26	0.24	0.23	0.22



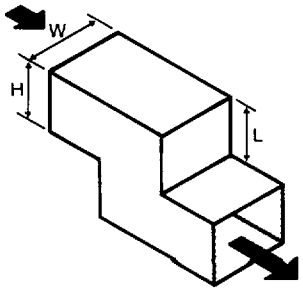




Loss Coefficients (C) for Double Thickness Vanes

Dimensions, Inches (mm)		Velocity, fpm (m/s)			
R	S	1000 (5)	1500 (7.5)	2000 (10)	2500 (12.5)
2.0 (50)	1.5 (38)	0.43	0.42	0.41	0.40
2.0 (50)	2.25 (56)	0.53	0.52	0.50	0.49
4.5 (114)	3.25 (83)	0.27	0.25	0.24	0.23

I. Elbows, 90°, Rectangular, Z-Shaped (15)



(NO VANES)

Coefficients for W/H = 1.0: (See Notes 4 and 5)

L/H	0	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
C	0	0.62	0.90	1.6	2.6	3.6	4.0	4.2	4.2	4.2
L/H	2.4	2.8	3.2	4.0	5.0	6.0	7.0	9.0	10.0	∞
C	3.7	3.3	3.2	3.1	2.9	2.8	2.7	2.6	2.5	2.3

Note 4: For W/H values other than 1.0 apply the following factor:

W/H	0.25	0.50	0.75	1.0	1.5	2.0	3.0	4.0	6.0	8.0
K	1.10	1.07	1.04	1.0	0.95	0.90	0.83	0.78	0.72	0.70

Note 5: Correction factor for Reynolds number — K_{Re}

R_e 10^{-4}	1	2	3	4	6	8	10	≥14
K_{Re}	1.40	1.26	1.19	1.14	1.09	1.06	1.04	1.0

U.S. Units
For Standard Air: $R_e = 8.56\ D\ V$

(Metric Units) ($R_e = 66.4\ DV$)

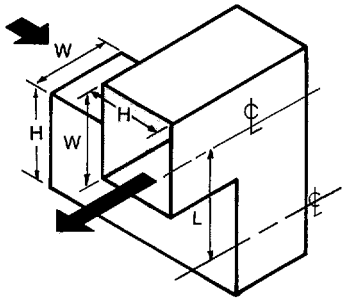
where:
D = duct diameter, inches (mm)
V = duct velocity, fpm (m/s)

For Rectangular Ducts:

$$D = \frac{2HW}{H + W}$$

Table 14-10 LOSS COEFFICIENTS, ELBOWS (Cont.)
Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

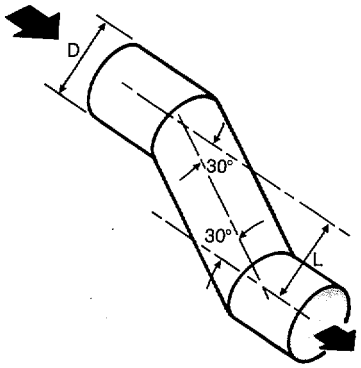
J. Elbows, 90°, Rectangular in Different Planes (15)
(See Chapter 5, Section H for new data on spin-in fittings)



Coefficients for H/W = 1.0: (See Notes 4 & 5—Page 14.22)

L/W	0	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
C	1.2	2.4	2.9	3.3	3.4	3.4	3.4	3.3	3.2	3.1
L/W	2.4	2.8	3.2	4.0	5.0	6.0	7.0	9.0	10.0	∞
C	3.2	3.2	3.2	3.0	2.9	2.8	2.7	2.5	2.4	2.3

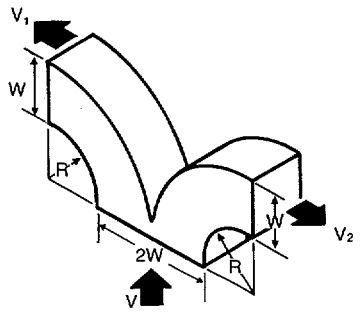
K. Elbows, 30°, Round, Offset (15)

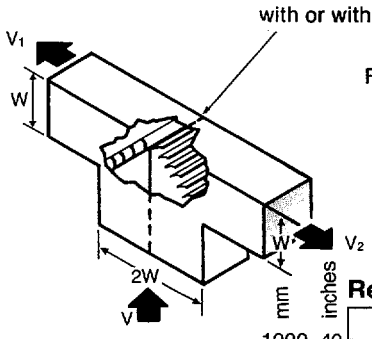


Coefficients C (See Note 5—Page 14.22)

L/D	0	0.5	1.0	1.5	2.0	2.5	3.0
C	0	0.15	0.15	0.16	0.16	0.16	0.16

L. Elbows, 90°, Rectangular Wye or Tee Shape





Flow can be diverging or converging

Use value of C for a similar elbow, provided $V = V_1 = V_2$ (Velocities)

Reynolds Number Correction Factor Chart

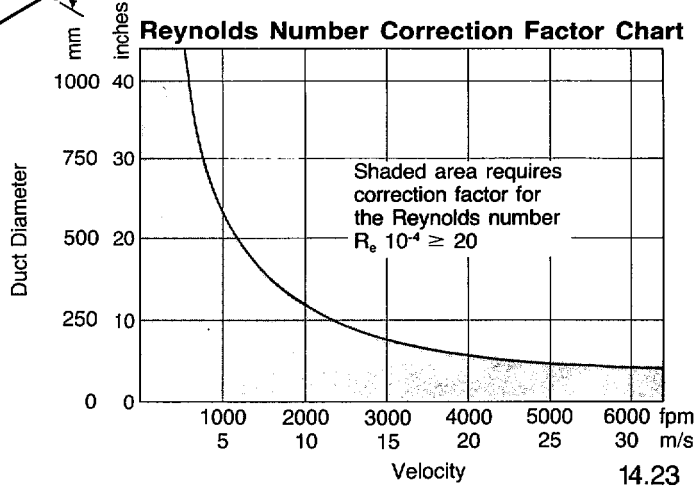
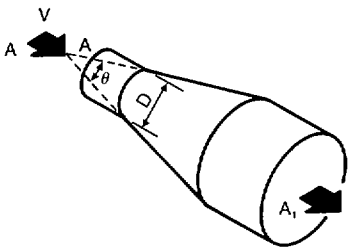
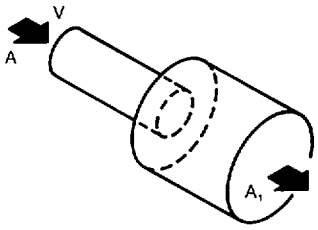


Table 14-11 LOSS COEFFICIENTS, TRANSITIONS (Diverging Flow)

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

A. Transition, Round, Conical (15)





When: θ = 180°

U.S. Units R_e = 8.56 DV **(Metric Units)** (R_e = 66.4 DV)

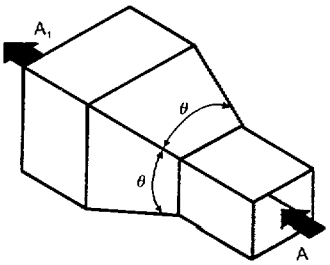
where:

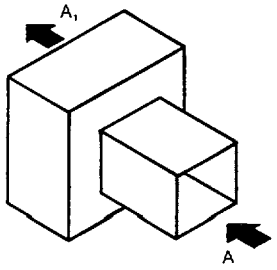
D = Upstream Diameter, inches (mm)
V = Upstream Velocity, fpm (m/s)

Coefficient C (See Note 6)

R_e	A_1/A	θ							
		16°	20°	30°	45°	60°	90°	120°	180°
0.5 × 10^5	2	0.14	0.19	0.32	0.33	0.33	0.32	0.31	0.30
	4	0.23	0.30	0.46	0.61	0.68	0.64	0.63	0.62
	6	0.27	0.33	0.48	0.66	0.77	0.74	0.73	0.72
	10	0.29	0.38	0.59	0.76	0.80	0.83	0.84	0.83
	≥16	0.31	0.38	0.60	0.84	0.88	0.88	0.88	0.88
2 × 10^5	2	0.07	0.12	0.23	0.28	0.27	0.27	0.27	0.26
	4	0.15	0.18	0.36	0.55	0.59	0.59	0.58	0.57
	6	0.19	0.28	0.44	0.90	0.70	0.71	0.71	0.69
	10	0.20	0.24	0.43	0.76	0.80	0.81	0.81	0.81
	≥16	0.21	0.28	0.52	0.76	0.87	0.87	0.87	0.87
≥6 × 10^5	2	0.05	0.07	0.12	0.27	0.27	0.27	0.27	0.27
	4	0.17	0.24	0.38	0.51	0.56	0.58	0.58	0.57
	6	0.16	0.29	0.46	0.60	0.69	0.71	0.70	0.70
	10	0.21	0.33	0.52	0.60	0.76	0.83	0.84	0.83
	≥16	0.21	0.34	0.56	0.72	0.79	0.85	0.87	0.89

B. Transition, Rectangular, Pyramidal (15)





When θ = 180°

Coefficient C (See Note 6)

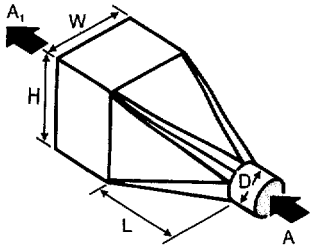
A_1/A	θ							
	16°	20°	30°	45°	60°	90°	120°	180°
2	0.18	0.22	0.25	0.29	0.31	0.32	0.33	0.30
4	0.36	0.43	0.50	0.56	0.61	0.63	0.63	0.63
6	0.42	0.47	0.58	0.68	0.72	0.76	0.76	0.75
≥10	0.42	0.49	0.59	0.70	0.80	0.87	0.85	0.86

Note 6: A = Area (Entering airstream), A_1 = Area (Leaving airstream)

Table 14-11 LOSS COEFFICIENTS, TRANSITIONS (Diverging Flow) (Cont.)

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

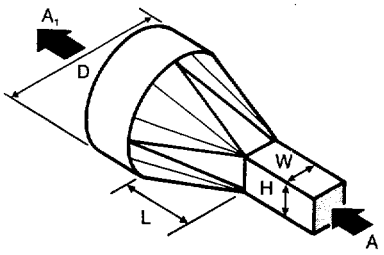
C. Transition, Round to Rectangular (15)



$$\tan\left(\frac{\theta}{2}\right) = \frac{1.13\sqrt{HW} - D}{2L}$$

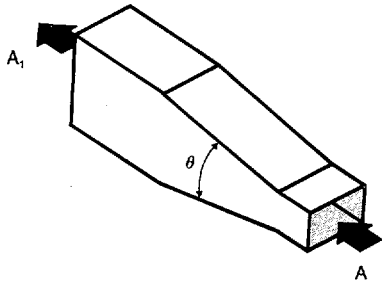
For Figures C and D, compute θ using the equations and find the coefficient C from Table 14-11B, Transition, Rectangular, Pyramidal.

D. Transition, Rectangular to Round (15)



$$\tan\left(\frac{\theta}{2}\right) = \frac{D - 1.13\sqrt{HW}}{2L}$$

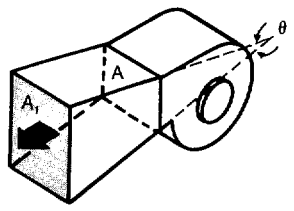
E. Transition, Rectangular, Sides Straight (15)



Coefficient C (See Note 6)

A_1/A	θ						
	14°	20°	30°	45°	60°	90°	180°
2	0.09	0.12	0.20	0.34	0.37	0.38	0.35
4	0.16	0.25	0.42	0.60	0.68	0.70	0.66
6	0.19	0.30	0.48	0.65	0.76	0.83	0.80

F. Transition, Symmetric at Fan With Duct Sides Straight (15)



Coefficient C (See Note 6)

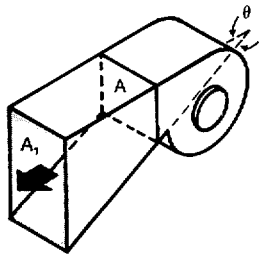
A_1/A	1.5	2.0	2.5	3.0	3.5	4.0
θ						
10°	0.05	0.07	0.09	0.10	0.11	0.11
15°	0.06	0.09	0.11	0.13	0.13	0.14
20°	0.07	0.10	0.13	0.15	0.16	0.16
25°	0.08	0.13	0.16	0.19	0.21	0.23
30°	0.16	0.24	0.29	0.32	0.34	0.35
35°	0.24	0.34	0.39	0.44	0.48	0.50

Note 6: A = Area (Entering airstream), A_1 = Area (Leaving airstream)

Table 14-11 LOSS COEFFICIENTS, TRANSITIONS (Diverging Flow) (Cont.)

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

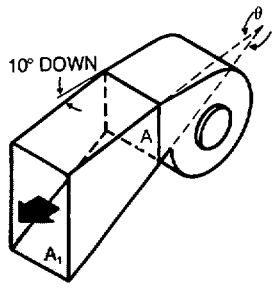
G. Transition, Asymmetric at Fan With Duct Sides Straight, Top Level (15)



Coefficient C (See Note 6)

A_1/A	1.5	2.0	2.5	3.0	3.5	4.0
θ						
10°	0.08	0.09	0.10	0.10	0.11	0.11
15°	0.10	0.11	0.12	0.13	0.14	0.15
20°	0.12	0.14	0.15	0.16	0.17	0.18
25°	0.15	0.18	0.21	0.23	0.25	0.26
30°	0.18	0.25	0.30	0.33	0.35	0.35
35°	0.21	0.31	0.38	0.41	0.43	0.44

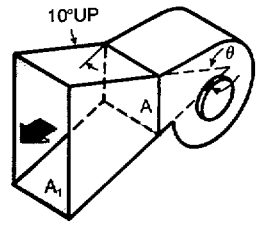
H. Transition, Asymmetric at Fan With Duct Sides Straight, Top 10° Down (15)



Coefficient C (See Note 6)

A_1/A	1.5	2.0	2.5	3.0	3.5	4.0
θ						
10°	0.11	0.13	0.14	0.14	0.14	0.14
15°	0.13	0.15	0.16	0.17	0.18	0.18
20°	0.19	0.22	0.24	0.26	0.28	0.30
25°	0.29	0.32	0.35	0.37	0.39	0.40
30°	0.36	0.42	0.46	0.49	0.51	0.51
35°	0.44	0.54	0.61	0.64	0.66	0.66

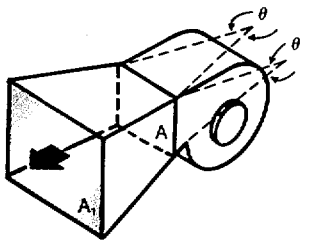
I. Transition, Asymmetric at Fan With Duct Sides Straight, Top 10° Up (15)



Coefficient C (See Note 6)

A_1/A	1.5	2.0	2.5	3.0	3.5	4.0
θ						
10°	0.05	0.08	0.11	0.13	0.13	0.14
15°	0.06	0.10	0.12	0.14	0.15	0.15
20°	0.07	0.11	0.14	0.15	0.16	0.16
25°	0.09	0.14	0.18	0.20	0.21	0.22
30°	0.13	0.18	0.23	0.26	0.28	0.29
35°	0.15	0.23	0.28	0.33	0.35	0.36

J. Transition, Pyramidal at Fan With Duct (15)



Coefficient C (See Note 6)

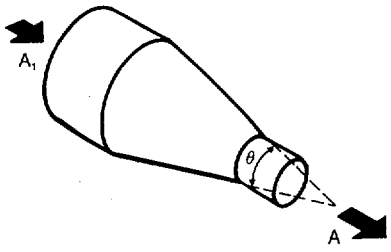
A_1/A	1.5	2.0	2.5	3.0	3.5	4.0
θ						
10°	0.10	0.18	0.21	0.23	0.24	0.25
15°	0.23	0.33	0.38	0.40	0.42	0.44
20°	0.31	0.43	0.48	0.53	0.56	0.58
25°	0.36	0.49	0.55	0.58	0.62	0.64
30°	0.42	0.53	0.59	0.64	0.67	0.69

Note 6: A = Area (Entering airstream), A_1 = Area (Leaving airstream)

Table 14-12 LOSS COEFFICIENTS, TRANSITIONS (Converging Flow)

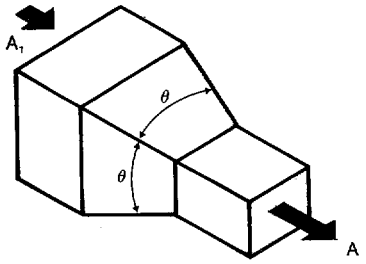
Use the velocity pressure (V_p) of the downstream section. Fitting loss (TP) = C × V_p

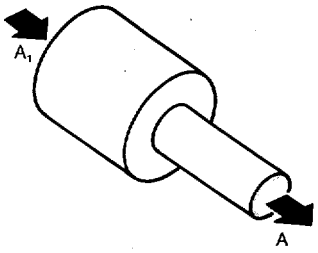
A. Contraction, Round and Rectangular, Gradual to Abrupt (15)



Coefficient C (See Note 7)

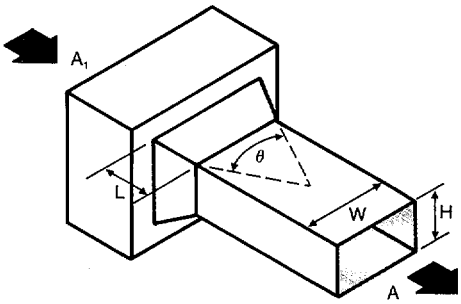
A_1/A	θ						
	10°	15°-40°	50°-60°	90°	120°	150°	180°
2	0.05	0.05	0.06	0.12	0.18	0.24	0.26
4	0.05	0.04	0.07	0.17	0.27	0.35	0.41
6	0.05	0.04	0.07	0.18	0.28	0.36	0.42
10	0.05	0.05	0.08	0.19	0.29	0.37	0.43





When θ = 180°

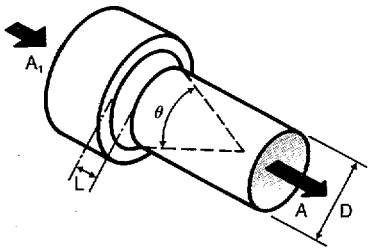
B. Contraction, Conical, Round and Rectangular (15)



For Rectangular: $D = \frac{2HW}{H+W}$

Coefficient C = K C_o

θ is major angle for rectangular transition



(See Note 7)

A/A_1	0	0.2	0.4	0.6	0.8	0.9	1.0
K	1.0	0.85	0.68	0.50	0.30	0.18	0

C_o

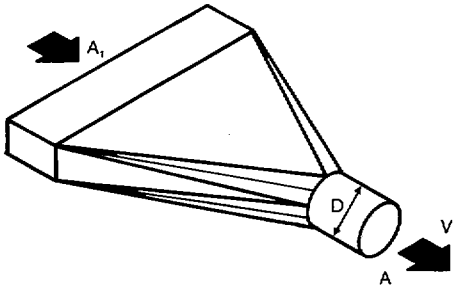
L/D	θ								
	0°	10°	20°	30°	40°	60°	100°	140°	180°
0.025	0.50	0.47	0.45	0.43	0.41	0.40	0.42	0.45	0.50
0.05	0.50	0.45	0.41	0.36	0.33	0.30	0.35	0.42	0.50
0.075	0.50	0.42	0.35	0.30	0.26	0.23	0.30	0.40	0.50
0.10	0.50	0.39	0.32	0.25	0.22	0.18	0.27	0.38	0.50
0.15	0.50	0.37	0.27	0.20	0.16	0.15	0.25	0.37	0.50
0.60	0.50	0.27	0.18	0.13	0.11	0.12	0.23	0.36	0.50

Note 7: A_1 = Area (Entering airstream), A = Area (Leaving airstream)

Table 14-12 LOSS COEFFICIENTS, TRANSITIONS (Converging Flow) (Cont.)

Use the velocity pressure (V_p) of the downstream section. Fitting loss (TP) = C × V_p

C. Contraction, Rectangular Slot to Round (15)



$A < A_1$

R_e = 8.56 DV (U.S. Units)
R_e = 66.4 DV (Metric Units)

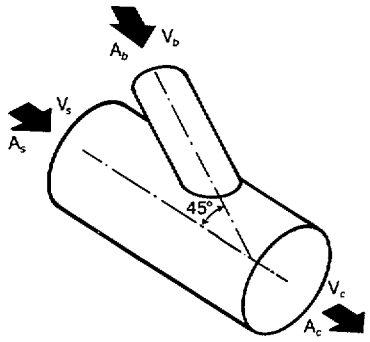
Coefficient C (See Note 7)

R_e 10^{-4}	1	2	4	6	8	10	20	≥40
C	0.27	0.25	0.20	0.17	0.14	0.11	0.04	0

Table 14-13 LOSS COEFFICIENTS, CONVERGING JUNCTIONS (Tees, Wyes)

Use the velocity pressure (V_p) of the downstream section. Fitting loss (TP) = C × V_p

A. Converging Wye, Round (2)



Branch, Coefficient C (See Note 8)

$\frac{V_b}{V_c}$	A_b/A_c						
	0.1	0.2	0.3	0.4	0.6	0.8	1.0
0.4	-.56	-.44	-.35	-.28	-.15	-.04	0.05
0.5	-.48	-.37	-.28	-.21	-.09	0.02	0.11
0.6	-.38	-.27	-.19	-.12	0	0.10	0.18
0.7	-.26	-.16	-.08	-.01	0.10	0.20	0.28
0.8	-.21	-.02	0.05	0.12	0.23	0.32	0.40
0.9	0.04	0.13	0.21	0.27	0.37	0.46	0.53
1.0	0.22	0.31	0.38	0.44	0.53	0.62	0.69
1.5	1.4	1.5	1.5	1.6	1.7	1.7	1.8
2.0	3.1	3.2	3.2	3.2	3.3	3.3	3.3
2.5	5.3	5.3	5.3	5.4	5.4	5.4	5.4
3.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0

Main, Coefficient C

$\frac{V_s}{V_c}$	A_b/A_c						
	0.1	0.2	0.3	0.4	0.6	0.8	1.0
0.1	-8.6	-4.1	-2.5	-1.7	-.97	-.58	-.34
0.2	-6.7	-3.1	-1.9	-1.3	-.67	-.36	-.18
0.3	-5.0	-2.2	-1.3	-.88	-.42	-.19	-.05
0.4	-3.5	-1.5	-.88	-.55	-.21	-.05	0.05
0.5	-2.3	-.95	-.51	-.28	-.06	0.06	0.13
0.6	-1.3	-.50	-.22	-.09	0.05	0.12	0.17
0.7	-.63	-.18	-.03	0.04	0.12	0.16	0.18
0.8	-.18	0.01	0.07	0.10	0.13	0.15	0.17
0.9	0.03	0.07	0.08	0.09	0.10	0.11	0.13
1.0	-0.01	0	0	0.10	0.02	0.04	0.05

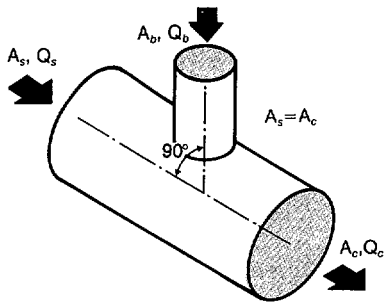
Note 7: A_1 = Area (Entering airstream), A = Area (Leaving airstream)

Note 8: A = Area, Q = Airflow, V = Velocity

Table 14-13 LOSS COEFFICIENTS, CONVERGING JUNCTIONS (Cont.)

Use the velocity pressure (V_p) of the downstream section. Fitting loss (TP) = C × V_p

B. Converging Tee, 90°, Round (15)



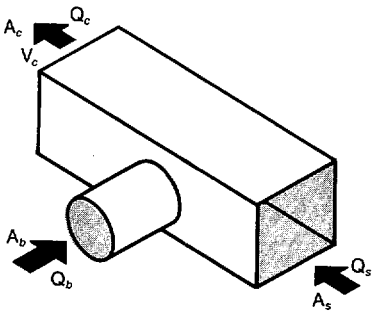
Branch, Coefficient C (See Note 8)

Q_b/Q_c	A_b/A_c						
	0.1	0.2	0.3	0.4	0.6	0.8	1.0
0.1	0.40	-.37	-.51	-.46	-.50	-.51	-.52
0.2	3.8	0.72	0.17	-.02	-.14	-.18	-.24
0.3	9.2	2.3	1.0	0.44	0.21	0.11	-.08
0.4	16	4.3	2.1	0.94	0.54	0.40	0.32
0.5	26	6.8	3.2	1.1	0.66	0.49	0.42
0.6	37	9.7	4.7	1.6	0.92	0.69	0.57
0.7	43	13	6.3	2.1	1.2	0.88	0.72
0.8	65	17	7.9	2.7	1.5	1.1	0.86
0.9	82	21	9.7	3.4	1.8	1.2	0.99
1.0	101	26	12	4.0	2.1	1.4	1.1

Main, Coefficient C (See Note 8)

Q_b/Q_c	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
C	0.16	0.27	0.38	0.46	0.53	0.57	0.59	0.60	0.59	0.55

C. Converging Tee, Round Branch to Rectangular Main



Branch, Coefficient C (See Note 8)

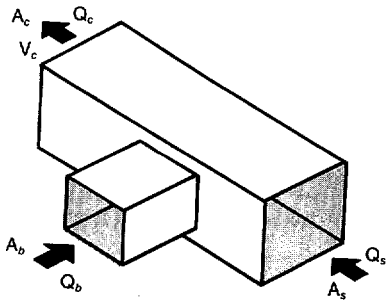
V_c fpm (m/s)	Q_b/Q_c									
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
< 1200 (6)	-.63	-.55	0.13	0.23	0.78	1.30	1.93	3.10	4.88	5.60
> 1200 (6)	-.49	-.21	0.23	0.60	1.27	2.06	2.75	3.70	4.93	5.95

For Main Loss Coefficients (C) see Fitting 14-13B

When:

A_b/A_s	A_s/A_c	A_b/A_c
0.5	1.0	0.5

D. Converging Tee, Rectangular Main and Branch



Branch, Coefficient (See Note 8)

V_c fpm (m/s)	Q_b/Q_c									
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
< 1200 (6)	-.75	-.53	-.03	0.33	1.03	1.10	2.15	2.93	4.18	4.78
> 1200 (6)	-.69	-.21	0.23	0.67	1.17	1.66	2.67	3.36	3.93	5.13

For Main Loss Coefficients (C) see Fitting 14-13B

When:

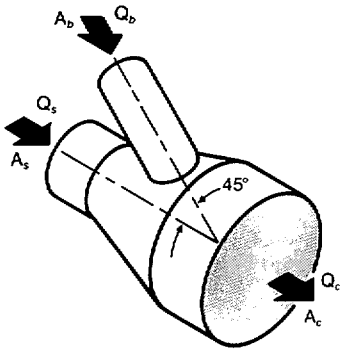
A_b/A_s	A_s/A_c	A_b/A_c
0.5	1.0	0.5

Note 8: A = Area, Q = Airflow, V = Velocity

Table 14-13 LOSS COEFFICIENTS, CONVERGING JUNCTIONS (Cont.)

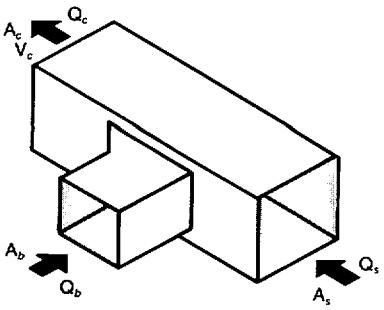
Use the velocity pressure (V_p) of the downstream section. Fitting loss (TP) = C × V_p

E. Converging Wye, Conical, Round (2)



$\frac{A_s}{A_c}$	$\frac{A_b}{A_c}$	Branch, Coefficient C (See Note 8) Q_b/Q_s									
		0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
0.3	0.2	-2.4	-.01	2.0	3.8	5.3	6.6	7.8	8.9	9.8	11
	0.3	-2.8	-1.2	0.12	1.1	1.9	2.6	3.2	3.7	4.2	4.6
0.4	0.2	-1.2	0.93	2.8	4.5	5.9	7.2	8.4	9.5	10	11
	0.3	-1.6	-.27	0.81	1.7	2.4	3.0	3.6	4.1	4.5	4.9
	0.4	-1.8	-.72	0.07	0.66	1.1	1.5	1.8	2.1	2.3	2.5
0.5	0.2	-.46	1.5	3.3	4.9	6.4	7.7	8.8	9.9	11	12
	0.3	-.94	0.25	1.2	2.0	2.7	3.3	3.8	4.2	4.7	5.0
	0.4	-1.1	-.24	0.42	0.92	1.3	1.6	1.9	2.1	2.3	2.5
	0.5	-1.2	-.38	0.18	0.58	0.88	1.1	1.3	1.5	1.6	1.7
0.6	0.2	-.55	1.3	3.1	4.7	6.1	7.4	8.6	9.6	11	12
	0.3	-1.1	0	0.88	1.6	2.3	2.8	3.3	3.7	4.1	4.5
	0.4	-1.2	-.48	0.10	0.54	0.89	1.2	1.4	1.6	1.8	2.0
	0.5	-1.3	-.62	-.14	0.21	0.47	0.68	0.85	0.99	1.1	1.2
	0.6	-1.3	-.69	-.26	0.04	0.26	0.42	0.57	0.66	0.75	0.82
0.8	0.2	0.06	1.8	3.5	5.1	6.5	7.8	8.9	10	11	12
	0.3	-.52	0.35	1.1	1.7	2.3	2.8	3.2	3.6	3.9	4.2
	0.4	-.67	-.05	0.43	0.80	1.1	1.4	1.6	1.8	1.9	2.1
	0.5	-.73	-.19	0.18	0.46	0.68	0.85	0.99	1.1	1.2	1.3
	0.6	-.75	-.27	0.05	0.28	0.45	0.58	0.68	0.76	0.83	0.88
	0.7	-.77	-.31	-.02	0.18	0.32	0.43	0.50	0.56	0.61	0.65
	0.8	-.78	-.34	-.07	0.12	0.24	0.33	0.39	0.44	0.47	0.50
1.0	0.2	—	2.1	3.7	5.2	6.6	7.8	9.0	11	11	12
	0.3	—	.54	1.2	1.8	2.3	2.7	3.1	3.7	3.7	4.0
	0.4	—	.21	0.62	0.96	1.2	1.5	1.7	2.0	2.0	2.1
	0.5	—	.05	0.37	0.60	0.79	0.93	1.1	1.2	1.2	1.3
	0.6	—	-.02	0.23	0.42	0.55	0.66	0.73	0.80	0.85	0.89
	0.8	—	-.10	0.11	0.24	0.33	0.39	0.43	0.46	0.47	0.48
	1.0	—	-.14	0.05	0.16	0.23	0.27	0.29	0.30	0.30	0.29

F. Converging Tee, 45° Entry Branch to Rectangular Main



When:

A_b/A_s	A_s/A_c	A_b/A_c
0.5	1.0	0.5

V_c fpm (m/s)	Branch, Coefficient C (See Note 8) Q_b/Q_c									
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
< 1200 (6)	-.83	-.68	-.30	0.28	0.55	1.03	1.50	1.93	2.50	3.03
> 1200 (6)	-.72	-.52	-.23	0.34	0.76	1.14	1.83	2.01	2.90	3.63

For Main Loss Coefficients (C) see Fitting 14-13B (Page 14.29)

Note 8: A = Area, Q = Airflow, V = Velocity

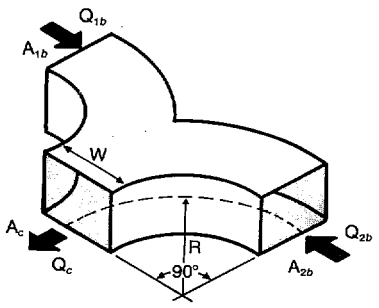
Table 14-13 LOSS COEFFICIENTS, CONVERGING JUNCTIONS (Cont.)

Use the velocity pressure (V_p) of the downstream section. Fitting loss (TP) = C × V_p

Main, Coefficient C (See Note 8)

A_s/A_c	A_b/A_c	Q_b/Q_s									
		0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
0.3	0.2	5.3	-.01	2.0	1.1	0.34	-.20	-.61	-.93	-1.2	-1.4
	0.3	5.4	3.7	2.5	1.6	1.0	0.53	0.16	-.14	-.38	-.58
0.4	0.2	1.9	1.1	0.46	-.07	-.49	-.83	-1.1	-1.3	-1.5	-1.7
	0.3	2.0	1.4	0.81	0.42	0.08	-.20	-.43	-.62	-.78	-.92
	0.4	2.0	1.5	1.0	0.68	0.39	0.16	-.04	-.21	-.35	-.47
0.5	0.2	0.77	0.34	-.09	-.48	-.81	-1.1	1.3	-1.5	-1.7	-1.8
	0.3	0.85	0.56	0.25	-.03	-.27	-.48	-.67	-.82	-.96	-1.1
	0.4	0.88	0.66	0.43	0.21	0.02	-.15	-.30	-.42	-.54	-.64
	0.5	0.91	0.73	0.54	0.36	0.21	0.06	-.06	-.17	-.26	-.35
0.6	0.2	0.30	0	-.34	-.67	-.96	-1.2	-1.4	-1.6	-1.8	-1.9
	0.3	0.37	0.21	-.02	-.24	-.44	-.63	-.79	-.93	-1.1	-1.2
	0.4	0.40	0.31	0.16	-.01	-.16	-.30	-.43	-.54	-.64	-.73
	0.5	0.43	0.37	0.26	0.14	0.02	-.09	-.20	-.29	-.37	-.45
	0.6	0.44	0.41	0.33	0.24	0.14	0.05	-.03	-.11	-.18	-.25
0.8	0.2	-.06	-.27	-.57	-.86	-1.1	-1.4	-1.6	-1.7	-1.9	-2.0
	0.3	0	-.08	-.25	-.43	-.62	-.78	-.93	-1.1	-1.2	-1.3
	0.4	0.04	0.02	-.08	-.21	-.34	-.46	-.57	-.67	-.77	-.85
	0.5	0.06	0.08	0.02	-.06	-.16	-.25	-.34	-.42	-.50	-.57
	0.6	0.07	0.12	0.09	0.03	-.04	-.11	-.18	-.25	-.31	-.37
	0.7	0.08	0.15	0.14	0.10	0.05	-.01	-.07	-.12	-.17	-.22
	0.8	0.09	0.17	0.18	0.16	0.11	0.07	0.02	-.02	-.07	-.11
1.0	0.2	—	-.39	-.67	-.96	-1.2	-1.5	-1.6	-1.8	-2.0	-2.1
	0.3	—	-.19	-.35	-.54	-.71	-.87	-1.0	-1.2	-1.3	-1.4
	0.4	—	-.10	-.19	-.31	-.43	-.55	-.66	-.77	-.86	-.94
	0.5	—	-.04	-.09	-.17	-.26	-.35	-.44	-.52	-.59	-.66
	0.6	—	0	-.02	-.07	-.14	-.21	-.28	-.34	-.40	-.46
	0.8	—	0.06	0.07	0.05	0.02	-.03	-.07	-.12	-.16	-.20
	1.0	—	0.09	0.13	0.13	0.11	0.08	0.06	0.03	-.01	-.03

G. Symmetrical Wye, Dovetail, Rectangular (15)



Coefficient C (See Note 8)

A_{1b}/A_c or A_{2b}/A_c	0.50	1.0
C	0.23	0.07

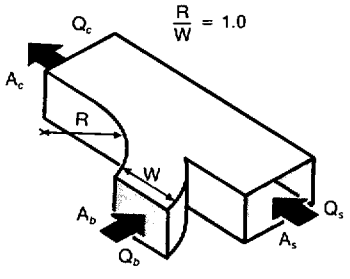
$\frac{R}{W} = 1.5 \qquad \frac{Q_{1b}}{Q_c} = \frac{Q_{2b}}{Q_c} = 0.5$

Note 8: A = Area, Q = Airflow, V = Velocity

Table 14-13 LOSS COEFFICIENTS, CONVERGING JUNCTIONS (Cont.)

Use the velocity pressure (V_p) of the downstream section. Fitting loss (TP) = C × V_p

H. Converging Wye, Rectangular (15)



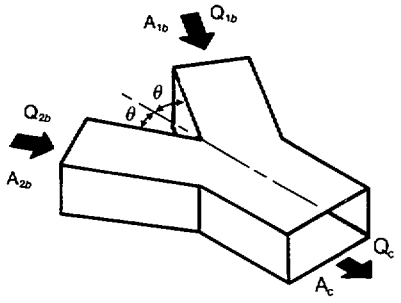
Branch, Coefficient (See Note 8)

A_b/A_s	A_b/A_c	Q_b/Q_c								
		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.25	0.25	-.50	0	0.50	1.2	2.2	3.7	5.8	8.4	11
0.33	0.25	-1.2	-.40	0.40	1.6	3.0	4.8	6.8	8.9	11
0.5	0.5	-.50	-.20	0	0.25	0.45	0.70	1.0	1.5	2.0
0.67	0.5	-1.0	-.60	-.20	0.10	0.30	0.60	1.0	1.5	2.0
1.0	0.5	-2.2	-1.5	-.95	-.50	0	0.40	0.80	1.3	1.9
1.0	1.0	-.60	-.30	-.10	-.04	0.13	0.21	0.29	0.36	0.42
1.33	1.0	-1.2	-.80	-.40	-.20	0	0.16	0.24	0.32	0.38
2.0	1.0	-2.1	-1.4	-.90	-.50	-.20	0	0.20	0.25	0.30

Main, Coefficient C (See Note 8)

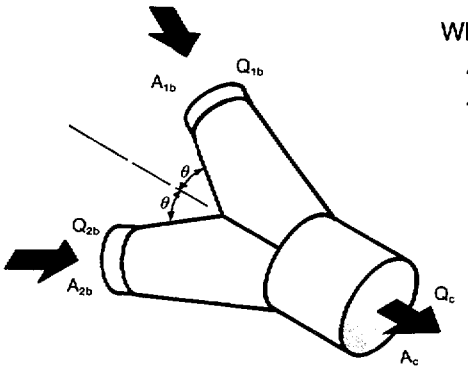
A_s/A_c	A_b/A_c	Q_b/Q_c								
		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.75	0.25	0.30	0.30	0.20	-.10	- .45	-.92	-1.5	-.2.0	-2.6
1.0	0.5	0.17	0.16	0.10	0	-0.08	-.18	-.27	-.37	-.46
0.75	0.5	0.27	0.35	0.32	0.25	0.12	-.03	-.23	-.42	-.58
0.5	0.5	1.2	1.1	0.90	0.65	0.35	0	-.40	-.80	-1.3
1.0	1.0	0.18	0.24	0.27	0.26	0.23	0.18	0.10	0	-.12
0.75	1.0	0.75	0.36	0.38	0.35	0.27	0.18	0.05	-.08	-.22
0.5	1.0	0.80	0.87	0.80	0.68	0.55	0.40	0.25	0.08	-.10

I. Wye, Rectangular and Round (15)



Coefficient C (See Note 8)

θ	Q_{1b}/Q_c or Q_{2b}/Q_c										
	0	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90	1.0
15°	-2.6	-1.9	-1.3	-.77	-.30	0.10	0.41	0.67	0.85	0.97	1.0
30°	-2.1	-1.5	-1.0	-.53	-.10	0.28	0.69	0.91	1.1	1.4	1.6
45°	-1.3	-.93	-.55	-.16	0.20	0.56	0.92	1.26	1.6	2.0	2.3



When:

$A_{1b} = A_{2b}$

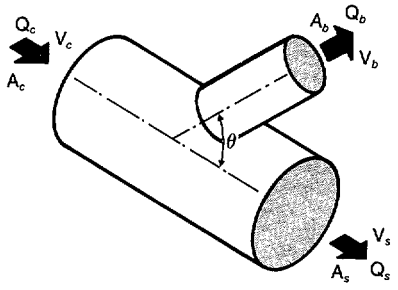
$A_c = A_{1b} + A_{2b}$

Note 8: A = Area, Q = Airflow, V = Velocity

Table 14-14 LOSS COEFFICIENTS, DIVERGING JUNCTIONS (Tees, Wyes)

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

A. Tee or Wye, 30° to 90°, Round (15)



Main, Coefficient C (See Note 8)

V_s/V_c	0	0.1	0.2	0.3	0.4	0.5	0.6	0.8	1.0
C	0.35	0.28	0.22	0.17	0.13	0.09	0.06	0.02	0

Wye θ = 30°: Branch, Coefficient C (See note 8)

A_b/A_c	Q_b/Q_c								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.8	0.75	0.55	0.40	0.28	0.21	0.16	0.15	0.16	0.19
0.7	0.72	0.51	0.36	0.25	0.18	0.15	0.16	0.20	0.26
0.6	0.69	0.46	0.31	0.21	0.17	0.16	0.20	0.28	0.39
0.5	0.65	0.41	0.26	0.19	0.18	0.22	0.32	0.47	0.67
0.4	0.59	0.33	0.21	0.20	0.27	0.40	0.62	0.92	1.3
0.3	0.55	0.28	0.24	0.38	0.76	1.3	2.0	—	—
0.2	0.40	0.26	0.58	1.3	2.5	—	—	—	—
0.1	0.28	1.5	—	—	—	—	—	—	—

Wye θ = 45°: Branch, Coefficient C (See note 8)

A_b/A_c	Q_bQ_c								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.8	0.78	0.62	0.49	0.40	0.34	0.31	0.32	0.35	0.40
0.7	0.77	0.59	0.47	0.38	0.34	0.32	0.35	0.41	0.50
0.6	0.74	0.56	0.44	0.37	0.35	0.36	0.43	0.54	0.68
0.5	0.71	0.52	0.41	0.38	0.40	0.45	0.59	0.78	1.0
0.4	0.66	0.47	0.40	0.43	0.54	0.69	0.95	1.3	1.7
0.3	0.66	0.48	0.52	0.73	1.2	1.8	2.7	—	—
0.2	0.56	0.56	1.0	1.8	—	—	—	—	—
0.1	0.60	2.1	—	—	—	—	—	—	—

Wye θ = 60°: Branch, Coefficient C (See note 8)

A_b/A_c	Q_b/Q_c								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.8	0.83	0.71	0.62	0.56	0.52	0.50	0.53	0.60	0.68
0.7	0.82	0.69	0.61	0.56	0.54	0.54	0.60	0.70	0.82
0.6	0.81	0.68	0.60	0.58	0.58	0.61	0.72	0.87	1.1
0.5	0.79	0.66	0.61	0.62	0.68	0.76	0.94	1.2	1.5
0.4	0.76	0.65	0.65	0.74	0.89	1.1	1.4	1.8	2.3
0.3	0.80	0.75	0.89	1.2	1.8	2.6	3.5	—	—
0.2	0.77	0.96	1.6	2.5	—	—	—	—	—
0.1	1.0	2.9	—	—	—	—	—	—	—

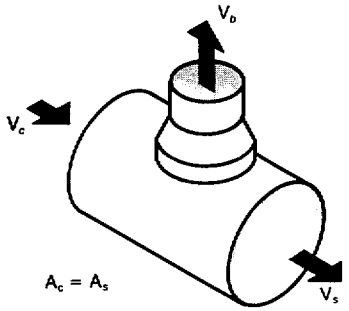
Tee θ = 90°: Branch, Coefficient C (See note 8)

A_b/A_c	Q_b/Q_c								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.8	0.95	0.92	0.92	0.93	0.94	0.95	1.1	1.2	1.4
0.7	0.95	0.94	0.95	0.98	1.0	1.1	1.2	1.4	1.6
0.6	0.96	0.97	1.0	1.1	1.1	1.2	1.4	1.7	2.0
0.5	0.97	1.0	1.1	1.2	1.4	1.5	1.8	2.1	2.5
0.4	0.99	1.1	1.3	1.5	1.7	2.0	2.4	—	—
0.3	1.1	1.4	1.8	2.3	—	—	—	—	—
0.2	1.3	1.9	2.9	—	—	—	—	—	—
0.1	2.1	—	—	—	—	—	—	—	—

Note 8: A = Area, Q = Airflow, V = Velocity

Table 14-14 LOSS COEFFICIENTS, DIVERGING JUNCTIONS (Cont.)
Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

B. 90° Conical Tee, Round (2)

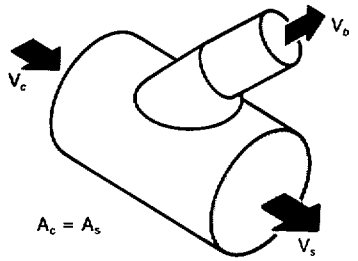


Branch, Coefficient C (See Note 8)

V_b/V_c	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
C	1.0	0.85	0.74	0.62	0.52	0.42	0.36	0.32	0.32	0.37	0.52

For Main Loss Coefficient (C) see Fitting 14-14A (Page 14.33)

C. 45° Conical Wye, Round (2)

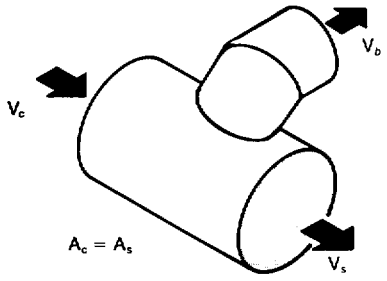


Branch, Coefficient C (See Note 8)

V_b/V_c	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
C	1.0	0.84	0.61	0.41	0.27	0.17	0.12	0.12	0.14	0.18	0.27

For Main Loss Coefficient (C) see Fitting 14-14A (Page 14.33)

D. 90° Tee, Round, Rolled 45° with 45° Elbow, Branch 90° to Main (2)

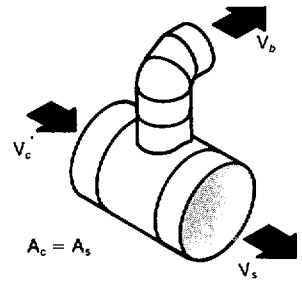


Branch, Coefficient C (See Note 8)

V_b/V_c	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
C	1.0	1.32	1.51	1.60	1.65	1.74	1.87	2.0	2.2	2.5	2.7

For Main Loss Coefficient (C) see Fitting 14-14A (Page 14.33)

E. 90° Tee, Round, with 90° Elbow, Branch 90° to Main (2)



Branch, Coefficient C (See Note 8)

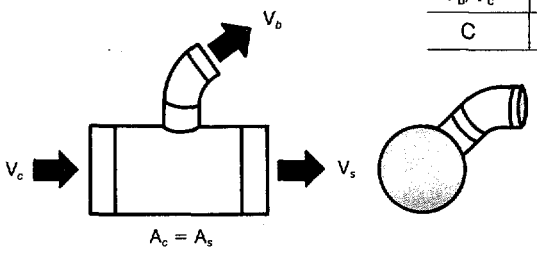
V_b/V_c	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
C	1.0	1.03	1.08	1.18	1.33	1.56	1.86	2.2	2.6	3.0	3.4

For Main Loss Coefficient (C) see Fitting 14-14A (Page 14.33)

Note 8: A = Area, Q = Airflow, V = Velocity

Table 14-14 LOSS COEFFICIENTS, DIVERGING JUNCTIONS (Cont.)
Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

F. 90° Tee, Round, Rolled 45° with 60° Elbow, Branch 45° to Main (2)

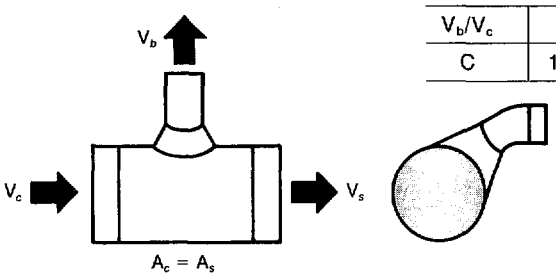


Branch, Coefficient C (See Note 8)

V_b/V_c	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
C	1.0	1.06	1.15	1.29	1.45	1.65	1.89	2.2	2.5	2.9	3.3

For Main Loss Coefficient (C) see Fitting 14-14A (Page 14.33)

G. 90° Conical Tee, Round, Rolled 45° with 45° Elbow, Branch 90° to Main (2)

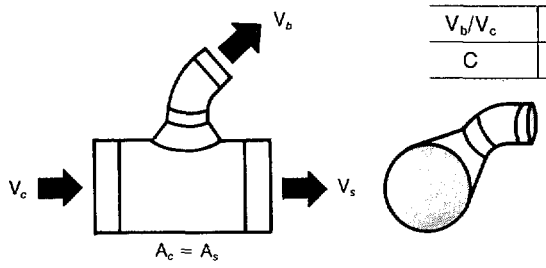


Branch, Coefficient C (See Note 8)

V_b/V_c	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
C	1.0	0.94	0.88	0.84	0.80	0.82	0.84	0.87	0.90	0.95	1.02

For Main Loss Coefficient (C) see Fitting 14-14A (Page 14.33)

H. 90° Conical Tee, Round, Rolled 45° with 60° Elbow, Branch 45° to Main (2)

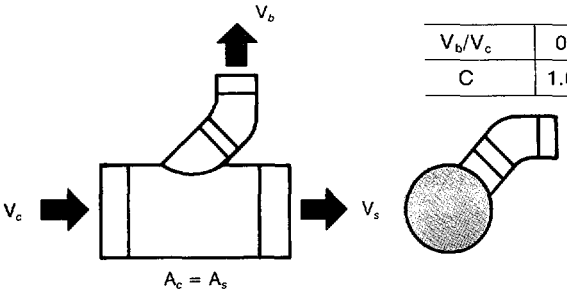


Branch, Coefficient C (See Note 8)

V_b/V_c	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
C	1.0	0.95	0.90	0.86	0.81	0.79	0.79	0.81	0.86	0.96	1.10

For Main Loss Coefficient (C) see Fitting 14-14A (Page 14.33)

I. 45° Wye, Round, Rolled 45° with 60° Elbow, Branch 90° to Main (2)



Branch, Coefficient C (See Note 8)

V_b/V_c	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
C	1.0	0.88	0.77	0.68	0.65	0.69	0.73	0.88	1.14	1.54	2.2

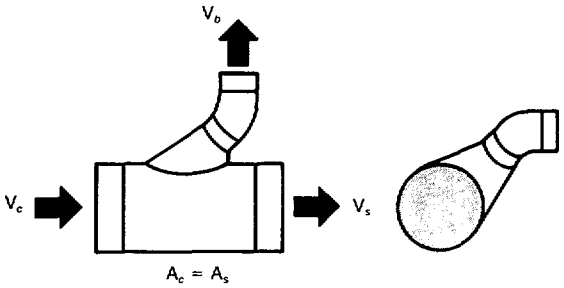
For Main Loss Coefficient (C) see Fitting 14-14A (Page 14.33)

Note 8: A = Area, Q = Airflow, V = Velocity

Table 14-14 LOSS COEFFICIENTS, DIVERGING JUNCTIONS (Cont.)

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

J. 45° Conical Wye, Round, Rolled 45° with 60° Elbow, Branch 90° to Main (2)

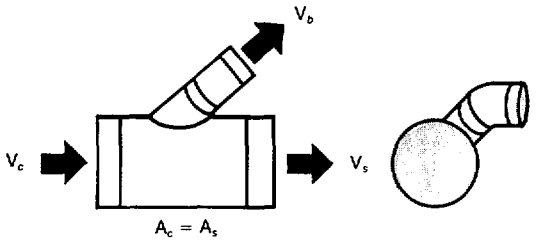


Branch, Coefficient C (See Note 8)

V_b/V_c	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
C	1.0	0.82	0.63	0.52	0.45	0.42	0.41	0.40	0.41	0.45	0.56

For Main Loss Coefficient (C) see Fitting 14-14A (Page 14.33)

K. 45° Wye, Round, Rolled 45° with 30° Elbow, Branch 45° to Main (2)

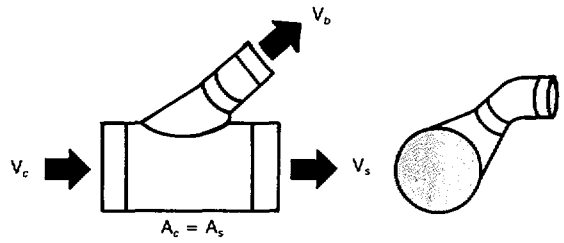


Branch, Coefficient C (See Note 8)

V_b/V_c	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
C	1.0	0.84	0.72	0.62	0.54	0.50	0.56	0.71	0.92	1.22	1.66

For Main Loss Coefficient (C) see Fitting 14-14A (Page 14.33)

L. 45° Conical Wye, Round, Rolled 45° with 30° Elbow, Branch 45° to Main (2)



Branch, Coefficient C (See Note 8)

V_b/V_c	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
C	1.0	0.93	0.71	0.55	0.44	0.42	0.42	0.44	0.47	0.54	0.62

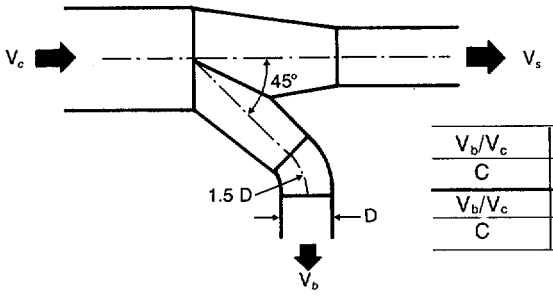
For Main Loss Coefficient (C) see Fitting 14-14A (Page 14.33)

Note 8: A = Area, Q = Airflow, V = Velocity

Table 14-14 LOSS COEFFICIENTS, DIVERGING JUNCTIONS (Cont.)

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

M. 45° Wye, Conical Main and Branch with 45° Elbow, Branch 90° to Main (15)



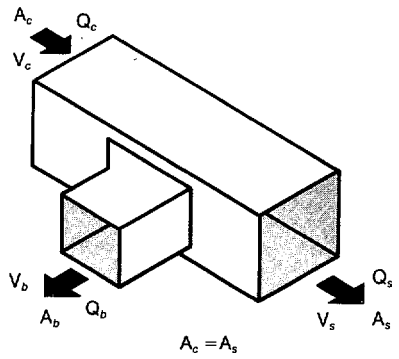
Branch, Coefficient C (See Note 8)

V_b/V_c	0.2	0.4	0.6	0.7	0.8	0.9	1.0	1.1	1.2
C	0.76	0.60	0.52	0.50	0.51	0.52	0.56	0.61	0.68
V_b/V_c	1.4	1.6	1.8	2.0	2.2	2.4	2.6	2.8	3.0
C	0.86	1.1	1.4	1.8	2.2	2.6	3.1	3.7	4.2

Main, Coefficient C (See Note 8)

V_s/V_c	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
C	0.14	0.06	0.05	0.09	0.18	0.30	0.46	0.64	0.84	1.0

N. Tee, 45° Entry, Rectangular Main and Branch

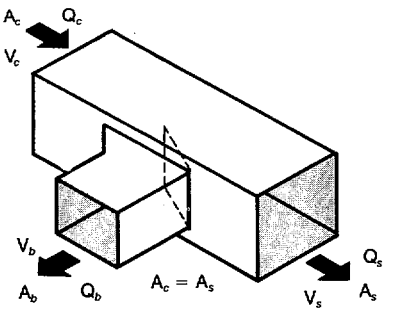


Branch, Coefficient C (See Note 8)

V_b/V_c	Q_b/Q_c								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.2	0.91								
0.4	0.81	0.79							
0.6	0.77	0.72	0.70						
0.8	0.78	0.73	0.69	0.66					
1.0	0.78	0.98	0.85	0.79	0.74				
1.2	0.90	1.11	1.16	1.23	1.03	0.86			
1.4	1.19	1.22	1.26	1.29	1.54	1.25	0.92		
1.6	1.35	1.42	1.55	1.59	1.63	1.50	1.31	1.09	
1.8	1.44	1.50	1.75	1.74	1.72	2.24	1.63	1.40	1.17

For Main Loss Coefficient (C) see Fitting 14-14A (Page 14.33)

P. Tee, 45° Entry, Rectangular Main and Branch with Damper



Branch, Coefficient C (See Note 8)

V_b/V_c	Q_b/Q_c								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.2	0.61								
0.4	0.46	0.61							
0.6	0.43	0.50	0.54						
0.8	0.39	0.43	0.62	0.53					
1.0	0.34	0.57	0.77	0.73	0.68				
1.2	0.37	0.64	0.85	0.98	1.07	0.83			
1.4	0.57	0.71	1.04	1.16	1.54	1.36	1.18		
1.6	0.89	1.08	1.28	1.30	1.69	2.09	1.81	1.47	
1.8	1.33	1.34	2.04	1.78	1.90	2.40	2.77	2.23	1.92

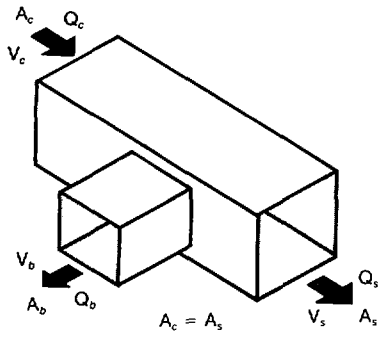
For Main Loss Coefficient (C) see Fitting 14-14S (Page 14.38)

Note 8: A = Area, Q = Airflow, V = Velocity

Table 14-14 LOSS COEFFICIENTS, DIVERGING JUNCTIONS (Cont.)

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

Q. Tee, Rectangular Main and Branch

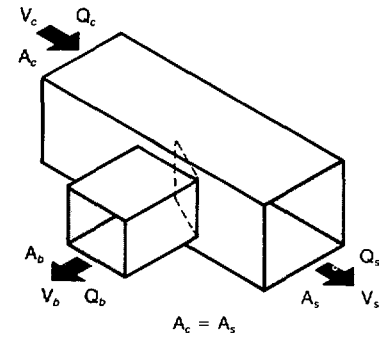


Branch, Coefficient C (See Note 8)

V_b/V_c	Q_b/Q_c								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.2	1.03								
0.4	1.04	1.01							
0.6	1.11	1.03	1.05						
0.8	1.16	1.21	1.17	1.12					
1.0	1.38	1.40	1.30	1.36	1.27				
1.2	1.52	1.61	1.68	1.91	1.47	1.66			
1.4	1.79	2.01	1.90	2.31	2.28	2.20	1.95		
1.6	2.07	2.28	2.13	2.71	2.99	2.81	2.09	2.20	
1.8	2.32	2.54	2.64	3.09	3.72	3.48	2.21	2.29	2.57

For Main Loss Coefficient (C) see Fitting 14-14A (Page 14.33)

R. Tee, Rectangular Main and Branch with Damper

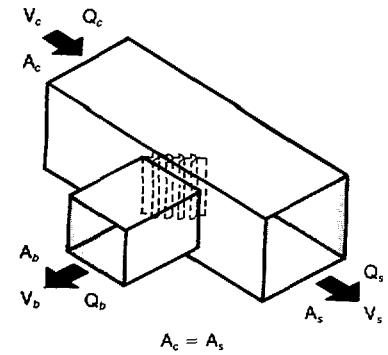


Branch, Coefficient C (See Note 8)

V_b/V_c	Q_b/Q_c								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.2	0.58								
0.4	0.67	0.64							
0.6	0.78	0.76	0.75						
0.8	0.88	0.98	0.81	1.01					
1.0	1.12	1.05	1.08	1.18	1.29				
1.2	1.49	1.48	1.40	1.51	1.70	1.91			
1.4	2.10	2.21	2.25	2.29	2.32	2.48	2.53		
1.6	2.72	3.30	2.84	3.09	3.30	3.19	3.29	3.16	
1.8	3.42	4.58	3.65	3.92	4.20	4.15	4.14	4.10	4.05

For Main Loss Coefficient (C) see Fitting 14-14S

S. Tee, Rectangular Main and Branch with Extractor



Branch, Coefficient C (See Note 8)

V_b/V_c	Q_b/Q_c								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.2	0.60								
0.4	0.62	0.69							
0.6	0.74	0.80	0.82						
0.8	0.99	1.10	0.95	0.90					
1.0	1.48	1.12	1.41	1.24	1.21				
1.2	1.91	1.33	1.43	1.52	1.55	1.64			
1.4	2.47	1.67	1.70	2.04	1.86	1.98	2.47		
1.6	3.17	2.40	2.33	2.53	2.31	2.51	3.13	3.25	
1.8	3.85	3.37	2.89	3.23	3.09	3.03	3.30	3.74	4.11

Main, Coefficient C (See Note 8)

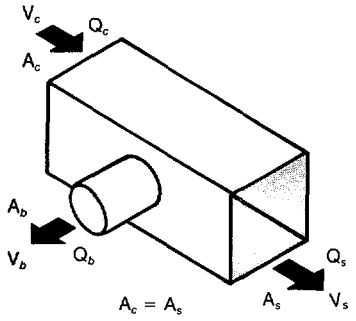
V_b/V_c	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8
C	0.03	0.04	0.07	0.12	0.13	0.14	0.27	0.30	0.25

Note 8: A = Area, Q = Airflow, V = Velocity

Table 14-14 LOSS COEFFICIENTS, DIVERGING JUNCTIONS (Cont.)

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

T. Tee, Rectangular Main to Round Branch

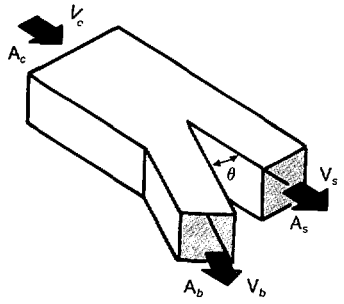


Branch, Coefficient C (See note 8)

V_b/V_c	Q_b/Q_c								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.2	1.00								
0.4	1.01	1.07							
0.6	1.14	1.10	1.08						
0.8	1.18	1.31	1.12	1.13					
1.0	1.30	1.38	1.20	1.23	1.26				
1.2	1.46	1.58	1.45	1.31	1.39	1.48			
1.4	1.70	1.82	1.65	1.51	1.56	1.64	1.71		
1.6	1.93	2.06	2.00	1.85	1.70	1.76	1.80	1.88	
1.8	2.06	2.17	2.20	2.13	2.06	1.98	1.99	2.00	2.07

For Main Loss Coefficient (C) see Fitting 14-14A (Page 14.33)

U. Wye, Rectangular (15)



θ = 15° to 90°
$A_c = A_s + A_b$

Branch, Coefficient C (See Note 8)

θ	V_b/V_c												
	0.1	0.2	0.3	0.4	0.5	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
15°	0.81	0.65	0.51	0.38	0.28	0.20	0.11	0.06	0.14	0.30	0.51	0.76	1.0
30°	0.84	0.69	0.56	0.44	0.34	0.26	0.19	0.15	0.15	0.30	0.51	0.76	1.0
45°	0.87	0.74	0.63	0.54	0.45	0.38	0.29	0.24	0.23	0.30	0.51	0.76	1.0
60°	0.90	0.82	0.79	0.66	0.59	0.53	0.43	0.36	0.33	0.39	0.51	0.76	1.0
90°	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0

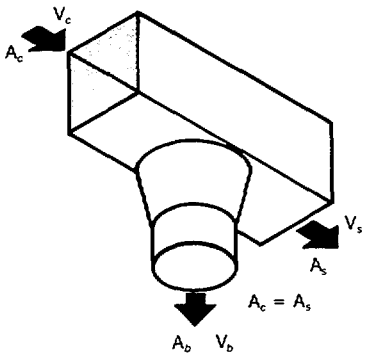
Main, Coefficient C (See Note 8)

θ	15°-60°	90°				
V_s/V_c	A_s/A_c					
	0-1.0	0-0.4	0.5	0.6	0.7	≥0.8
0	1.0	1.0	1.0	1.0	1.0	1.0
0.1	0.81	0.81	0.81	0.81	0.81	0.81
0.2	0.64	0.64	0.64	0.64	0.64	0.64
0.3	0.50	0.50	0.52	0.52	0.50	0.50
0.4	0.36	0.36	0.40	0.38	0.37	0.36
0.5	0.25	0.25	0.30	0.28	0.27	0.25
0.6	0.16	0.16	0.23	0.20	0.18	0.16
0.8	0.04	0.04	0.17	0.10	0.07	0.04
1.0	0	0	0.20	0.10	0.05	0
1.2	0.07	0.07	0.36	0.21	0.14	0.07
1.4	0.39	0.39	0.79	0.59	0.39	—
1.6	0.90	0.90	1.4	1.2	—	—
1.8	1.8	1.8	2.4	—	—	—
2.0	3.2	3.2	4.0	—	—	—

Note 8: A = Area, Q = Airflow, V = Velocity

Table 14-14 LOSS COEFFICIENTS, DIVERGING JUNCTIONS (Cont.)
Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

V. Tee Rectangular Main to Conical Branch (2)

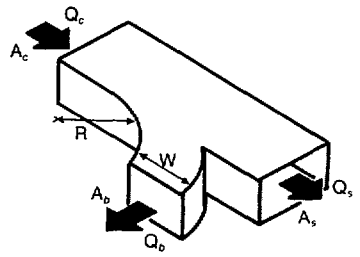


Branch, Coefficient C (See note 8)

V_b/V_c	0.40	0.50	0.75	1.0	1.3	1.5
C	0.80	0.83	0.90	1.0	1.1	1.4

For Main Loss Coefficient (C) see Fitting 14-14A (Page 14.33)

W. Wye, Rectangular (15)



$\frac{R}{W} = 1.0$

90° Branch

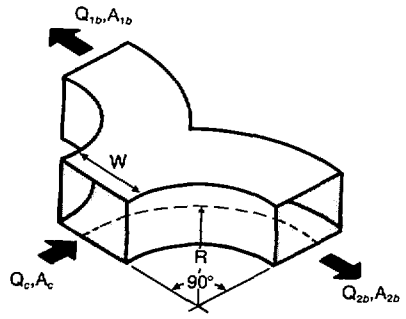
Branch, Coefficient C (See Note 8)

A_b/A_s	A_b/A_c	Q_b/Q_c								
		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.25	0.25	0.55	0.50	0.60	0.85	1.2	1.8	3.1	4.4	6.0
0.33	0.25	0.35	0.35	0.50	0.80	1.3	2.0	2.8	3.8	5.0
0.5	0.5	0.62	0.48	0.40	0.40	0.48	0.60	0.78	1.1	1.5
0.67	0.5	0.52	0.40	0.32	0.30	0.34	0.44	0.62	0.92	1.4
1.0	0.5	0.44	0.38	0.38	0.41	0.52	0.68	0.92	1.2	1.6
1.0	1.0	0.67	0.55	0.46	0.37	0.32	0.29	0.29	0.30	0.37
1.33	1.0	0.70	0.60	0.51	0.42	0.34	0.28	0.26	0.26	0.29
2.0	1.0	0.60	0.52	0.43	0.33	0.24	0.17	0.15	0.17	0.21

Main, Coefficient C (See Note 8)

A_b/A_s	A_b/A_c	Q_b/Q_c								
		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.25	0.25	-.01	-.03	-.01	0.05	0.13	0.21	0.29	0.38	0.46
0.33	0.25	0.08	0	-.02	-.01	0.02	0.08	0.16	0.24	0.34
0.5	0.5	-.03	-.06	-.05	0	0.06	0.12	0.19	0.27	0.35
0.67	0.5	0.04	-.02	-.04	-.03	-.01	0.04	0.12	0.23	0.37
1.0	0.5	0.72	0.48	0.28	0.13	0.05	0.04	0.09	0.18	0.30
1.0	1.0	-.02	-.04	-.04	-.01	0.06	0.13	0.22	0.30	0.38
1.33	1.0	0.10	0	0.01	-.03	-.01	0.03	0.10	0.20	0.30
2.0	1.0	0.62	0.38	0.23	0.13	0.08	0.05	0.06	0.10	0.20

X. Symmetrical Wye, Dovetail, Rectangular (15)



Coefficient C (See Note 8)

A_{1b}/A_c or A_{2b}/A_c	0.50	1.0
C	0.30	0.25

When: R/W = 1.5

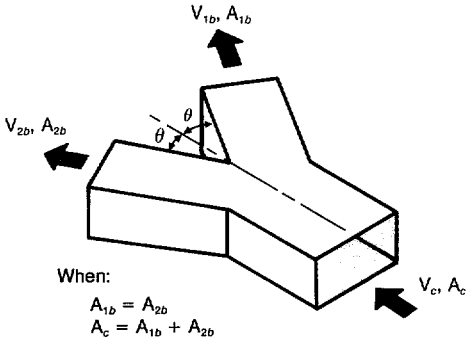
$$\frac{Q_{1b}}{Q_c} = \frac{Q_{2b}}{Q_c} = 0.5$$

Note 8: A = Area, Q = Airflow, V = Velocity

Table 14-14 LOSS COEFFICIENTS, DIVERGING JUNCTIONS (Cont.)

Use the velocity pressure (V^p) of the upstream section. Fitting loss (TP) = C × V_p

Y. Wye, Rectangular and Round (15)

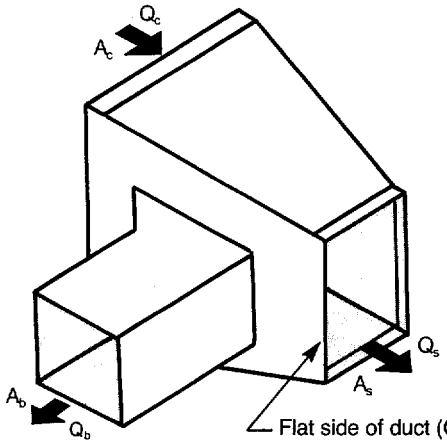


Coefficient C (See Note 8)

θ	V_{1b}/V_c or V_{2b}/V_c						
	0.1	0.2	0.3	0.4	0.5	0.6	0.8
15°	0.81	0.65	0.51	0.38	0.28	0.20	0.11
30°	0.84	0.69	0.56	0.44	0.34	0.26	0.19
45°	0.87	0.74	0.63	0.54	0.45	0.38	0.29
60°	0.90	0.82	0.79	0.66	0.59	0.53	0.43
90°	1.0	1.0	1.0	1.0	1.0	1.0	1.0

θ	V_{1b}/V_c or V_{2b}/V_c					
	1.0	1.2	1.4	1.6	1.8	2.0
15°	0.06	0.14	0.30	0.51	0.76	1.0
30°	0.15	0.15	0.30	0.51	0.76	1.0
45°	0.24	0.23	0.30	0.51	0.76	1.0
60°	0.36	0.33	0.39	0.51	0.76	1.0
90°	1.0	1.0	1.0	1.0	1.0	1.0

Z. Tee, Rectangular Reducing, 45° Entry Branch (2)



Branch, Coefficient C

Q_b/Q_c	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
C	1.16	0.96	0.82	0.68	0.56	0.49	0.47	0.48	0.50	0.54

Main, Coefficient C

Q_s/Q_c	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
C	0.21	0.20	0.20	0.20	0.20	0.20	0.22	0.25	0.35	0.53

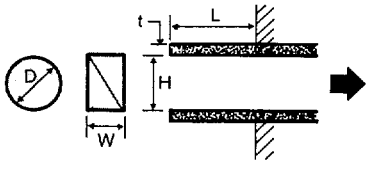
When:

A_b/A_c = 0.5 to 1.0
A_s/A_c = 0.5 to 1.0

Table 14-15 LOSS COEFFICIENTS, ENTRIES

Use the velocity pressure (V_p) of the downstream section. Fitting loss (TP) = C × V_p

A. Duct Mounted in Wall, Round and Rectangular (15)



Rectangular: $D = \frac{2\,HW}{(H + W)}$

Coefficient C

t/D	L/D						
	0	0.002	0.01	0.05	0.2	0.5	≥1.0
≈ 0	0.50	0.57	0.68	0.80	0.92	1.0	1.0
0.02	0.50	0.51	0.52	0.55	0.66	0.72	0.72
≥0.05	0.50	0.50	0.50	0.50	0.50	0.50	0.50

With Screen or Perforated Plate:

a. Sharp Edge ($t/D_e \leq 0.05$): $C_s = 1 + C_1$

b. Thick Edge ($t/D_e > 0.05$): $C_s = C + C_1$

where:

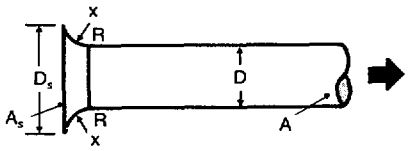
C_s is new coefficient of fitting with a screen or perforated plate at the entrance.
C is from above table
C_1 is from Table 14-17A (screen) or Table 14-17B (perforated plate)

Note 8: A = Area, Q = Airflow, V = Velocity

Table 14-15 LOSS COEFFICIENTS, ENTRIES (Cont.)

Use the velocity pressure (V_p) of the downstream section. Fitting loss (TP) = C × V_p

B. Smooth Converging Bellmouth, Round, without End Wall (15)

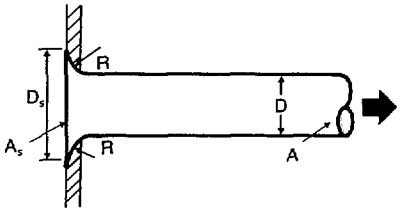


Coefficient C (See Note 9)

R/D	0	0.01	0.02	0.03	0.04	0.05
C	1.0	0.87	0.74	0.61	0.51	0.40

R/D	0.06	0.08	0.10	0.12	0.16	0.20 & greater
C	0.32	0.20	0.15	0.10	0.06	0.03

C. Smooth Converging Bellmouth, Round, with End Wall (15)

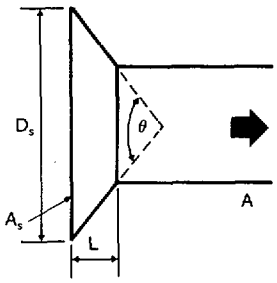


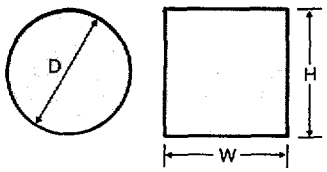
Coefficient C (See Note 9)

R/D	0	0.01	0.02	0.03	0.04	0.05
C	0.50	0.43	0.36	0.31	0.26	0.22

R/D	0.06	0.08	0.10	0.12	0.16	0.20 & greater
C	0.20	0.15	0.12	0.09	0.06	0.03

D. Conical, Converging Bellmouth, Round and Rectangular without End Wall (15)





Rectangular: $D = \frac{2\,HW}{(H + W)}$

θ is major angle for rectangular entry

Coefficient C (See Note 9)

L/D	θ 0°	10°	20°	30°	40°	60°	100°	140°	180°
0.025	1.0	0.96	0.93	0.90	0.86	0.80	0.69	0.59	0.50
0.05	1.0	0.93	0.86	0.80	0.75	0.67	0.58	0.53	0.50
0.10	1.0	0.80	0.67	0.55	0.48	0.41	0.41	0.44	0.50
0.25	1.0	0.68	0.45	0.30	0.22	0.17	0.22	0.34	0.50
0.60	1.0	0.46	0.27	0.18	0.14	0.13	0.21	0.33	0.50
1.0	1.0	0.32	0.20	0.14	0.11	0.10	0.18	0.30	0.50

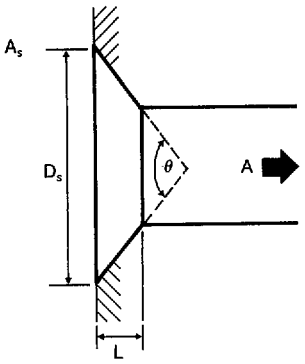
Note 9: With screen in opening at D_S, C_S = C (from table) + $\frac{\text{C (Screen coef. Table 14-17)}}{\left(\frac{A_S}{A}\right)^2}$

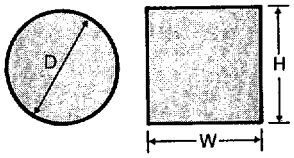
where: A= Area at D; A_S = Area at D_S

Table 14-15 LOSS COEFFICIENTS, ENTRIES (Cont.)

Use the velocity pressure (V_p) of the downstream section. Fitting loss (TP) = C × V_p

E. Conical, Converging Bellmouth, Round and Rectangular, with End Wall (15)





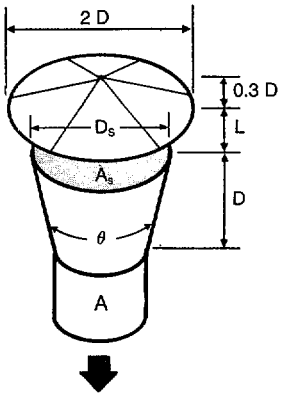
Rectangular: $D = \frac{2\,HW}{(H + W)}$

θ is major angle for rectangular entry

Coefficient C (See Note 9)

L/D	θ								
	0°	10°	20°	30°	40°	60°	100°	140°	180°
0.025	0.50	0.47	0.45	0.43	0.41	0.40	0.42	0.45	0.50
0.05	0.50	0.45	0.41	0.36	0.33	0.30	0.35	0.42	0.50
0.075	0.50	0.42	0.35	0.30	0.26	0.23	0.30	0.40	0.50
0.10	0.50	0.39	0.32	0.25	0.22	0.18	0.27	0.38	0.50
0.15	0.50	0.37	0.27	0.20	0.16	0.15	0.25	0.37	0.50
0.60	0.50	0.27	0.18	0.13	0.11	0.12	0.23	0.36	0.50

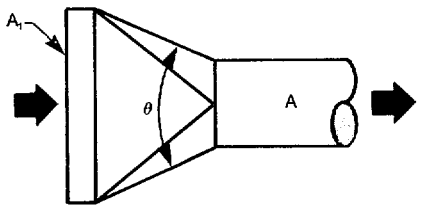
F. Intake Hood (15)



Coefficient C (See Note 9)

θ	L/D								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9 or greater
0°	2.6	1.8	1.5	1.4	1.3	1.2	1.2	1.1	1.1
15°	1.3	0.77	0.60	0.48	0.41	0.30	0.29	0.28	0.25

G. Hood, Tapered, Flanged or Unflanged (2)



$A_1 \geq 2A$

θ is major angle for rectangular hoods

Coefficient C

θ	0°	20°	40°	60°	80°	100°	120°	140°	160°	180°
Round Hood	1.0	0.11	0.06	0.09	0.14	0.18	0.27	0.32	0.43	0.50
Square Hood or Rect. Hood	1.0	0.19	0.13	0.16	0.21	0.27	0.33	0.43	0.53	0.62

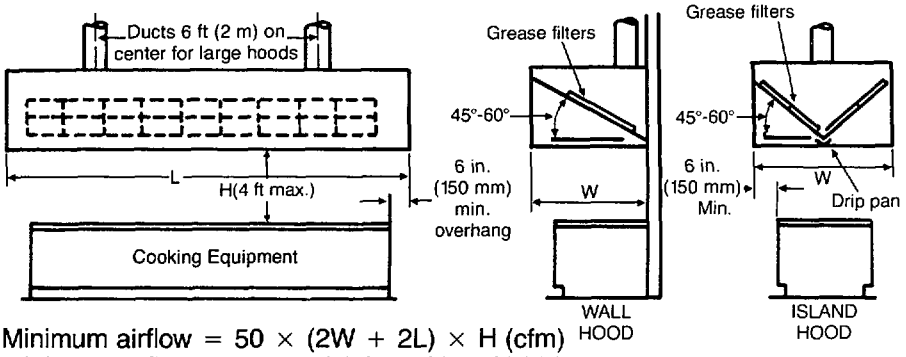
Note 9: With screen in opening at D_S, C_S = C (from table) + $\frac{C \text{ (Screen coef. Table 14-17)}}{\left(\frac{A_S}{A}\right)^2}$

where: A= Area at D; A_S = Area at D_S

Table 14-15 LOSS COEFFICIENTS, ENTRIES (Cont.)

Use the velocity pressure (V_p) of the downstream section. Fitting loss (TP) = C × V_p

H. Hood, Canopy Island or Range

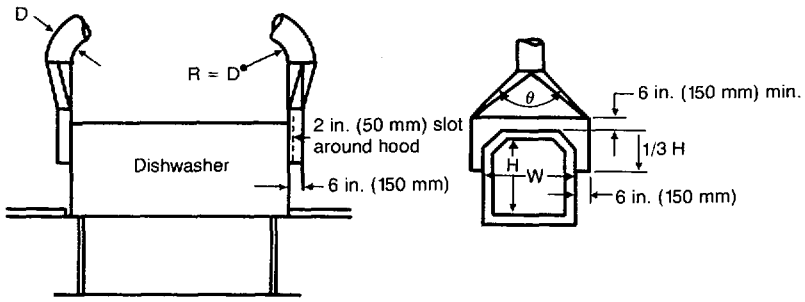


C = 0.5

Total Entry Loss = C × V_p
+ clean filter resistance + 0.1 in. w.g. (25 Pa)

Minimum airflow = 50 × (2W + 2L) × H (cfm)
Minimum airflow = 254 × (2W + 2L) × H (l/s)
Minimum face velocity = 125 fpm (0.62 m/s)
Minimum duct velocity = 1000 fpm (5 m/s)
(Use minimums from local codes where applicable)
W, L, H in feet (U.S.), in metres (metric)

I. Hood, Slot (Dishwasher)



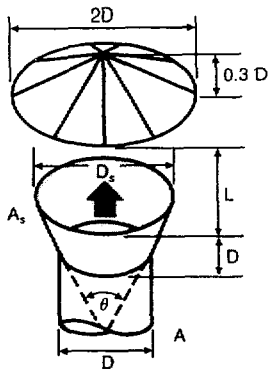
Use "C" loss coefficient from above Table 14-15 G. "Hood—Tapered, Flanged or Unflanged".
(Total Entry Loss = 1.0 × Slot V_p + C × Duct V_p)

Minimum airflow = 300 × H × W (cfm) = 1524 × H × W (l/s)
Minimum duct velocity = 1000 fpm (5 m/s)
(Use minimums from local codes where applicable)

Table 14-16 LOSS COEFFICIENTS, EXITS

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

A. Exhaust Hood (15)

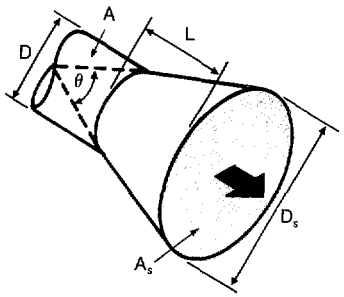


	Coefficient C (See Note 9)									
θ	L/D									
	0.1	0.2	0.25	0.3	0.35	0.4	0.5	0.6	0.8	1.0
0°	4.0	2.3	1.9	1.6	1.4	1.3	1.2	1.1	1.0	1.0
15°	2.6	1.2	1.0	0.80	0.70	0.65	0.60	0.60	0.60	0.60

Note 9: With screen in opening at D_S, C_S = C (from table) + $\frac{\text{C (Screen coef. Table 14-17)}}{\left(\frac{A_S}{A}\right)^2}$
where: A= Area at D; A_S = Area at D_S

Table 14-16 LOSS COEFFICIENTS, EXITS (Cont.)
Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

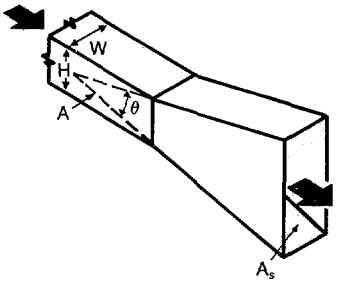
B. Exit, Conical, Round, with or without a Wall (15)



Coefficient C (See Note 9)

A_s/A	θ						
	14°	16°	20°	30°	45°	60°	≥90°
2	0.33	0.36	0.44	0.74	0.97	0.99	1.0
4	0.24	0.28	0.36	0.54	0.94	1.0	1.0
6	0.22	0.25	0.32	0.49	0.94	0.98	1.0
10	0.19	0.23	0.30	0.50	0.94	0.72	1.0
16	0.17	0.20	0.27	0.49	0.94	1.0	1.0

C. Exit, Plane Diffuser, Rectangular, with or without a Wall (15)

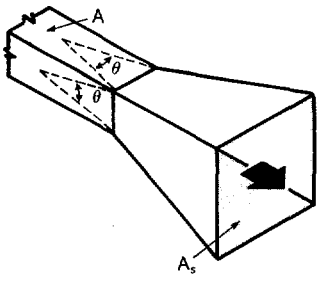


Coefficient C (See Note 10)

A_s/A	θ					
	14°	20°	30°	45°	60°	≥90°
2	0.37	0.38	0.50	0.75	0.90	1.1
4	0.25	0.37	0.57	0.82	1.0	1.1
6	0.28	0.47	0.64	0.87	1.0	1.1

When: 0.5 ≤ H/W ≤ 2.0

D. Exit, Pyramidal Diffuser, Rectangular, with or without a Wall (15)



Coefficient C (See Note 10)

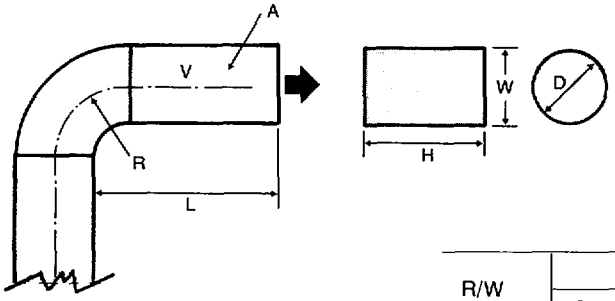
A_s/A	θ					
	10°	14°	20°	30°	45°	≥60°
2	0.44	0.58	0.70	0.86	1.0	1.1
4	0.31	0.48	0.61	0.76	0.94	1.1
6	0.29	0.47	0.62	0.74	0.94	1.1
10	0.26	0.45	0.60	0.73	0.89	1.0

Note 10: With screen in opening at A_s, C_s = C (from table) + $\frac{\text{C (Screen coef. Table 14-17)}}{\left(\frac{A_s}{A}\right)^2}$

Table 14-16 LOSS COEFFICIENTS, EXITS (Cont.)

Use the velocity pressure (V^p) of the upstream section. Fitting loss (TP) = C × V_p

E. Exit, Discharge to Atmosphere from a 90° Elbow, Round and Rectangular (15)



Note:
Elbow loss included in loss coefficient.

RECTANGULAR: Coefficient C (See Note 11)

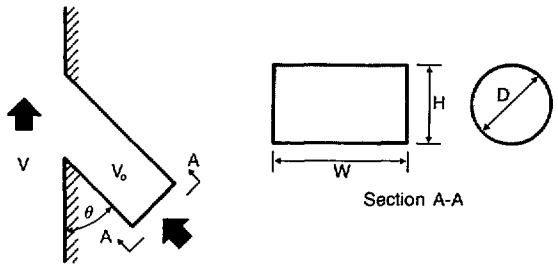
R/W	L/W									
	0	0.5	1.0	1.5	2.0	3.0	4.0	6.0	8.0	12.0
0	3.0	3.1	3.2	3.0	2.7	2.4	2.2	2.1	2.1	2.0
0.75	2.2	2.2	2.1	1.8	1.7	1.6	1.6	1.5	1.5	1.5
1.0	1.8	1.5	1.4	1.4	1.3	1.3	1.2	1.2	1.2	1.2
1.5	1.5	1.2	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
2.5	1.2	1.1	1.1	1.0	1.0	1.0	1.0	1.0	1.0	1.0

ROUND: Coefficient C (See Note 10)

L/D	0.9	1.3
C	1.5	1.4

When:
R/D = 1.0 (Round)

F. Exit, Duct Flush with Wall, Flow along Wall (15)



RECTANGULAR:
Coefficient C (See Note 11)

Aspect Ratio (H/W)	θ	V/V_o				
		0	0.5	1.0	1.5	2.0
0.1-0.2	30°-90°	1.0	0.95	1.2	1.5	1.8
	120°	1.0	0.92	1.1	1.4	1.9
	150°	1.0	0.75	0.95	1.4	1.8
0.5-2.0	30°-45°	1.0	1.0	1.1	1.3	1.6
	60°	1.0	0.90	1.1	1.4	1.6
	90°	1.0	0.80	0.95	1.4	1.7
	120°	1.0	0.80	0.95	1.3	1.7
	150°	1.0	0.82	0.83	1.0	1.3
5-10	45°	1.0	0.92	0.93	1.1	1.3
	60°	1.0	0.87	0.87	1.0	1.3
	90°	1.0	0.82	0.80	0.97	1.2
	120°	1.0	0.80	0.76	0.90	0.98

ROUND: Coefficient C (See Note 10)

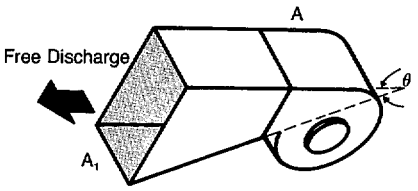
θ	V/V_o				
	0	0.5	1.0	1.5	2.0
30°-45°	1.0	1.0	1.1	1.3	1.6
60°	1.0	0.90	1.1	1.4	1.6
90°	1.0	0.80	0.95	1.4	1.7
120°	1.0	0.80	0.95	1.3	1.7
150°	1.0	0.82	0.83	1.0	1.3

NOTE 11: With screen in opening C_s = C (from table) + C (Screen Coef. from Table 14-17)

Table 14-16 LOSS COEFFICIENTS, EXITS (Cont.)

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

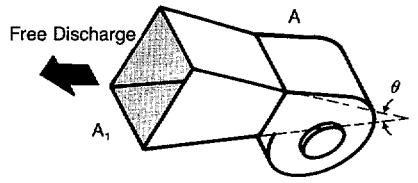
G. Plane Asymmetric Diffuser at Fan Outlet without Ductwork (15)



Coefficient C (See Note 12)

θ	A_1/A					
	1.5	2.0	2.5	3.0	3.5	4.0
10°	0.51	0.34	0.25	0.21	0.18	0.17
15°	0.54	0.36	0.27	0.24	0.22	0.20
20°	0.55	0.38	0.31	0.27	0.25	0.24
25°	0.59	0.43	0.37	0.35	0.33	0.33
30°	0.63	0.50	0.46	0.44	0.43	0.42
35°	0.65	0.56	0.53	0.52	0.51	0.50

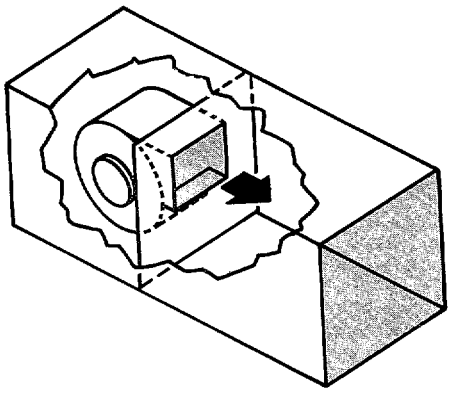
H. Pyramidal Diffuser at Fan Outlet without Ductwork (15)



Coefficient C (See Note 12)

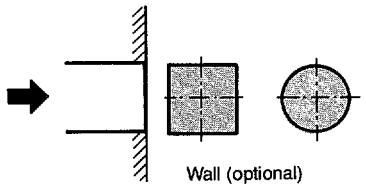
θ	A_1/A					
	1.5	2.0	2.5	3.0	3.5	4.0
10°	0.54	0.42	0.37	0.34	0.32	0.31
15°	0.67	0.58	0.53	0.51	0.50	0.51
20°	0.75	0.67	0.65	0.64	0.64	0.65
25°	0.80	0.74	0.72	0.70	0.70	0.72
30°	0.85	0.78	0.76	0.75	0.75	0.76

I. Fan, Free Discharge, Plenum



C = 1.0 (Airfoil wheels—Backwardly inclined)
C = 1.5 (Backwardly inclined wheels)
C = 1.8 (Forward curved wheels)

J. Exit, Abrupt, Round and Rectangular, with or without a Wall (15)



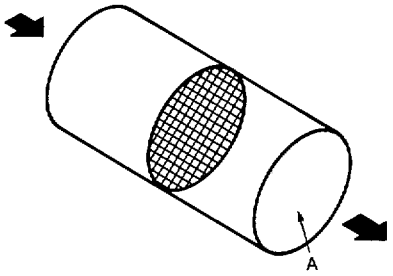
C = 1.0
With Screen: C_s = 1 + (C from Table 14-17)

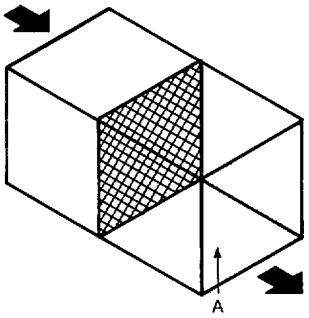
NOTE 12: With screen in opening at A_1, C_s = C (from table) + $C \frac{\text{(Screen coef. Table 14-17)}}{\left(\frac{A_1}{A}\right)^2}$

Table 14-17 LOSS COEFFICIENTS, SCREENS AND PLATES

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

A. Screen in Duct, Round and Rectangular (15)





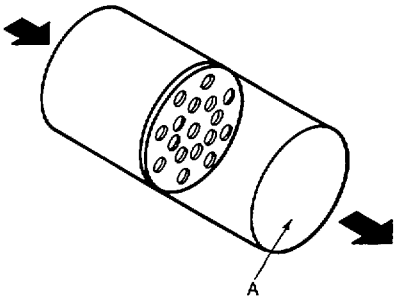
$n = A_s/A$

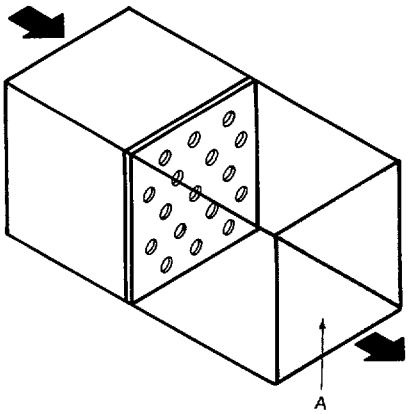
Where:
- n = Free area ratio of screen
- A_s = Total flow area of screen
- A = Area of Duct

Coefficient C

n	0.30	0.40	0.50	0.55	0.60	0.65	0.70	0.75	0.80	0.90	1.0
C	6.2	3.0	1.7	1.3	0.97	0.75	0.58	0.44	0.32	0.14	0

B. Perforated Plate in Duct, Thick, Round and Rectangular (15)





Coefficient C

t/d	n								
	0.20	0.25	0.30	0.40	0.50	0.60	0.70	0.80	0.90
0.015	52	30	18	8.2	4.0	2.0	0.97	0.42	0.13
0.2	48	28	17	7.7	3.8	1.9	0.91	0.40	0.13
0.4	46	27	17	7.4	3.6	1.8	0.88	0.39	0.13
0.6	42	24	15	6.6	3.2	1.6	0.80	0.36	0.13

t/d > 0.015 Where:

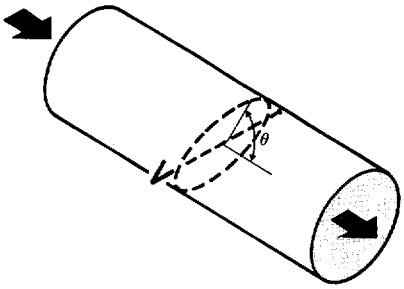
$$n = \frac{A_p}{A}$$

- t = plate thickness
- d = diameter of perforated holes
- n = free area ratio of plate
- A_p = total flow area of perforated plate
- A = area of duct

Table 14-18 LOSS COEFFICIENTS, OBSTRUCTIONS (Constant Velocities)

Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

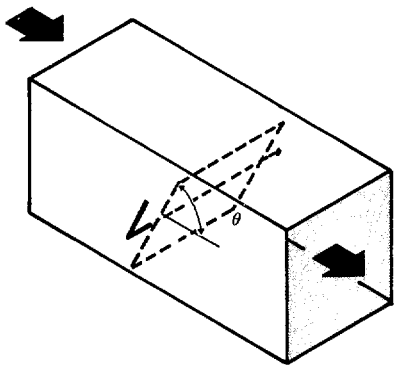
A. Damper, Butterfly, Thin Plate, Round (15)



	Coefficient C						
θ	0°	10°	20°	30°	40°	50°	60°
C	0.20	0.52	1.5	4.5	11	29	108

0° is full open

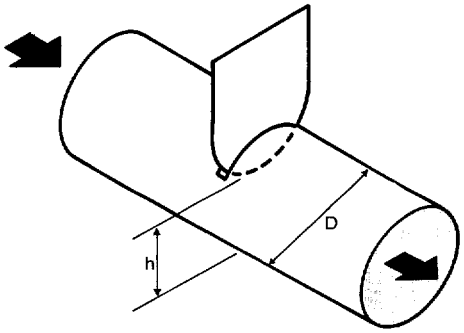
B. Damper, Butterfly, Thin Plate, Rectangular (15)



	Coefficient C						
θ	0°	10°	20°	30°	40°	50°	60°
C	0.04	0.33	1.2	3.3	9.0	26	70

0° is full open

C. Damper, Gate, Round (15)



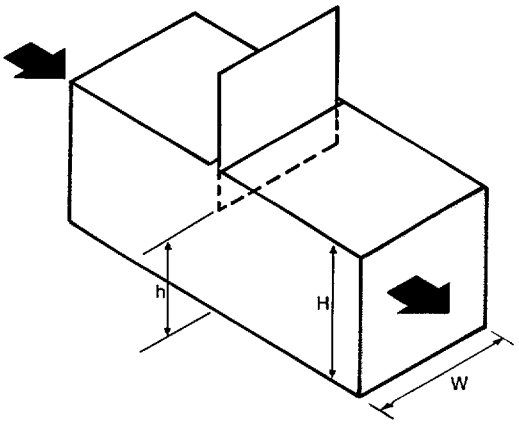
	Coefficient C							
h/D	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
A_h/A	0.25	0.38	0.50	0.61	0.71	0.81	0.90	0.96
C	35	10	4.6	2.1	0.98	0.44	0.17	0.06

A_h = Free Area
A = Area of Duct

Table 14-18 LOSS COEFFICIENTS, OBSTRUCTIONS (Constant Velocities) (Cont.)

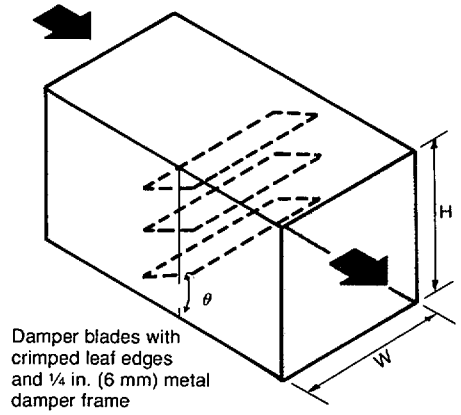
Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

D. Damper, Gate, Rectangular (15)



H/W	Coefficient C h/H						
	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.5	14	6.9	3.3	1.7	0.83	0.32	0.09
1.0	19	8.8	4.5	2.4	1.2	0.55	0.17
1.5	20	9.1	4.7	2.7	1.2	0.47	0.11
2.0	18	8.8	4.5	2.3	1.1	0.51	0.13

E. Damper, Rectangular, Parallel Blades (2)



Damper blades with crimped leaf edges and ¼ in. (6 mm) metal damper frame

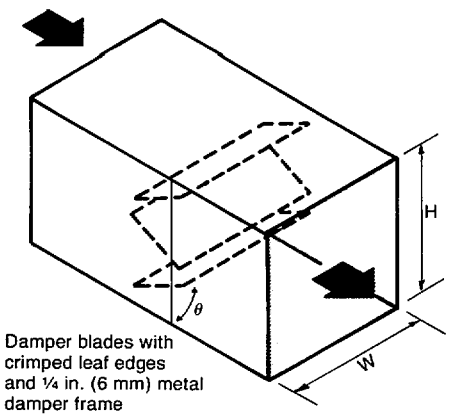
L/R	Coefficient C θ								
	80°	70°	60°	50°	40°	30°	20°	10°	0° Fully open
0.3	116	32	14	9.0	5.0	2.3	1.4	0.79	0.52
0.4	152	38	16	9.0	6.0	2.4	1.5	0.85	0.52
0.5	188	45	18	9.0	6.0	2.4	1.5	0.92	0.52
0.6	245	45	21	9.0	5.4	2.4	1.5	0.92	0.52
0.8	284	55	22	9.0	5.4	2.5	1.5	0.92	0.52
1.0	361	65	24	10	5.4	2.6	1.6	1.0	0.52
1.5	576	102	28	10	5.4	2.7	1.6	1.0	0.52

$$\frac{L}{R} = \frac{N\,W}{2\,(H + W)}$$

where:

N is number of damper blades
W is duct dimension parallel to blade axis
L is sum of damper blade lengths
R is perimeter of duct
H is duct dimension perpendicular to blade axis

F. Damper, Rectangular, Opposed Blades (2)



Damper blades with crimped leaf edges and ¼ in. (6 mm) metal damper frame

L/R	Coefficient C θ								
	80°	70°	60°	50°	40°	30°	20°	10°	0° Fully open
0.3	807	284	73	21	9.0	4.1	2.1	0.85	0.52
0.4	915	332	100	28	11	5.0	2.2	0.92	0.52
0.5	1045	377	122	33	13	5.4	2.3	1.0	0.52
0.6	1121	411	148	38	14	6.0	2.3	1.0	0.52
0.8	1299	495	188	54	18	6.6	2.4	1.1	0.52
1.0	1521	547	245	65	21	7.3	2.7	1.2	0.52
1.5	1654	677	361	107	28	9.0	3.2	1.4	0.52

$$\frac{L}{R} = \frac{N\,W}{2\,(H + W)}$$

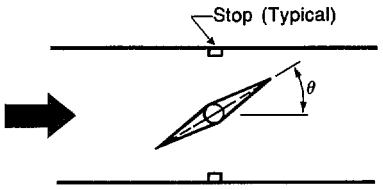
where:

N is number of damper blades
W is duct dimension parallel to blade axis
L is sum of damper blade lengths
R is perimeter of duct
H is duct dimension perpendicular to blade axis

Table 14-18 LOSS COEFFICIENTS, OBSTRUCTIONS (Constant Velocities) (Cont.)

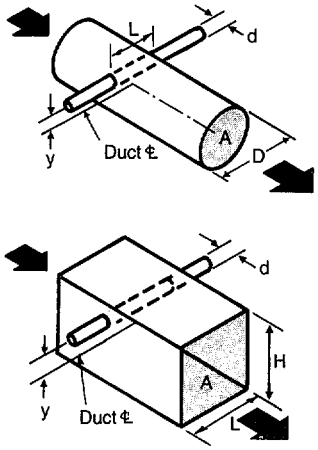
Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

G. Damper, Butterfly, Airfoil Blade, Rectangular (15)



	Coefficient C						
θ	0°	10°	20°	30°	40°	50°	60°
C	0.50	0.65	1.6	4.0	9.4	24	67

H. Obstruction, Smooth Cylinder in Round and Rectangular Ducts (15)



	Coefficient C_o			
R_e	S_m/A			
	.05	.10	.15	.20
0. 1	3.9	8.4	14	19
0. 5	1.5	3.2	5.2	7.1
1	0.66	1.4	2.3	3.2
5	0.30	0.64	1.1	1.4
10	0.17	0.38	0.62	0.84
50	0.11	0.24	0.38	0.52
100	0.10	0.21	0.35	0.47
(0.5-200) × 10^3	0.07	0.15	0.24	0.33
3 × 10^5	0.07	0.16	0.26	0.35
4 × 10^5	0.05	0.11	0.19	0.25
5 × 10^5	0.04	0.09	0.14	0.19
(6-10) × 10^5	0.02	0.05	0.07	0.10

where:

S_m = d L

R_e = 8.56 DV (Std. Air—U.S.)

R_e = 66.4 DV (Std. Air-Metric)

$S_m/A < 0.3$

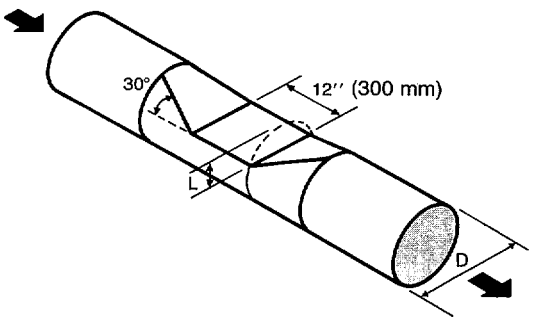
C = K C_o

A = Area of duct

For obstruction offset from the centerline use the following factors:

y/D or y/H	0	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40
K	1.0	0.97	0.93	0.89	0.84	0.79	0.74	0.67	0.58

I. Round Duct, Depressed to Avoid an Obstruction

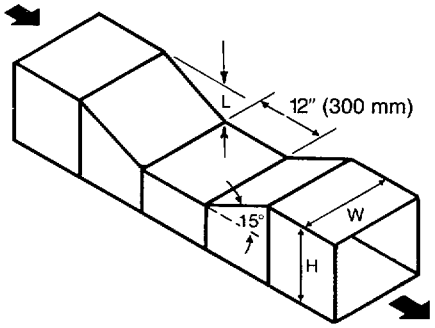


When: L/D = 0.33

C = 0.24

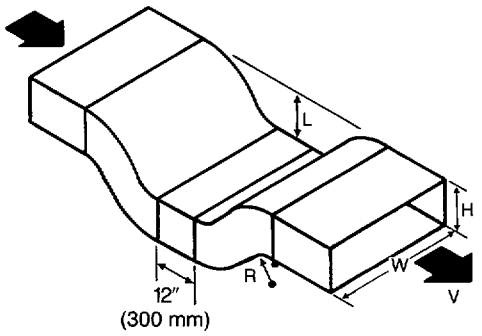
Table 14-18 LOSS COEFFICIENTS, OBSTRUCTIONS (Constant Velocities) (Cont.)
Use the velocity pressure (V_p) of the upstream section. Fitting loss (TP) = C × V_p

J. Rectangular Duct, Depressed to Avoid an Obstruction



Coefficient C				
W/H	L/H			
	0.125	0.15	0.25	0.30
1.0	0.26	0.30	0.33	0.35
4.0	0.10	0.14	0.22	0.30

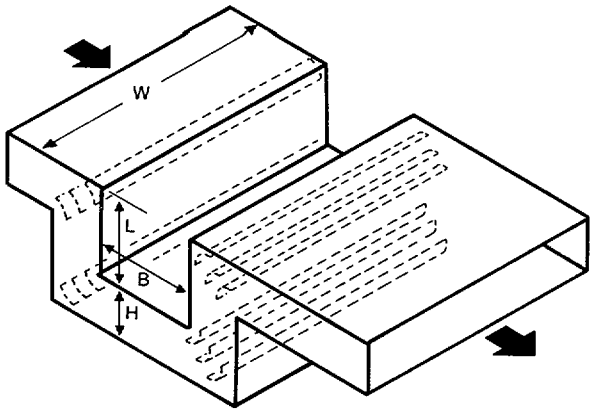
K. Rectangular Duct with 4-45° Smooth Radius Ells to Avoid an Obstruction



Coefficient C					
Velocity fpm (m/s)	800 (4)	1200 (6)	1600 (8)	2000 (10)	2400 (12)
C	0.18	0.22	0.24	0.25	0.26

Where:
W/H = 4
R/H = 1
L/H = 1.5

L. Rectangular Duct with 4-90° Mitered Ells to Avoid an Obstruction



Coefficient C				
L/H Ratio	0.5	1.0	1.5	2
Single Blade Turning Vanes	—	0.86	0.83	0.77
Double Blade Turning Vanes	—	1.85	2.84	2.91
"S" type Splitter Vanes	0.61	0.65	—	—
No Vanes—Up to 1200 fpm (6 m/s)	0.88	5.26	6.92	7.56
No Vanes—Over 1200 fpm (6 m/s)	1.26	6.22	8.82	9.24

Where: W/H = 1.0 to 3.0
B = 12" to 24"